

STAGNANT WATER AND WATERBORNE DISEASES: IMPLICATIONS FOR CHILDREN'S HEALTH AND EDUCATION*

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Abstract

Waterborne diseases cause over 6 billion diarrheal episodes annually and remain a leading cause of child mortality. Here, we develop a novel measure of stagnant water, a key driver of these diseases, and estimate its impact on Tanzanian children's health, mortality, and education. We find that a 10 percentage point increase in stagnant water raises diarrhea incidence by 30%, and significantly increases child mortality. Stagnant water also reduces cognitive performance and raises school absenteeism. Climate change is expected to dramatically impact the burden from stagnant water shocks due to joint effects of higher rainfall intensity and temperatures. These findings highlight the importance of targeted policies to mitigate stagnant water risks, especially in vulnerable urban settings.

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I. Introduction

Waterborne diseases are the second most common type of disease in the world and lead to more than 6 billion diarrheal episodes per year (Murray et al., 2020), with most of the health burden on children in low-income countries (Prüss et al., 2002).¹ Despite the persistently high prevalence of waterborne diseases, and the increasing evidence on long-term effects of childhood health shocks (Currie and Almond, 2011), we have limited knowledge on the consequences of childhood exposure to waterborne diseases. The aim of this paper is to evaluate the causal effect of exposure to an important risk factor of waterborne diseases: stagnant water, on child health, mortality, and educational outcomes.

Waterborne diseases spread through physical contact with pathogen-contaminated food or water that infects the gastrointestinal system.² Stagnant water enables the spread of waterborne pathogens through the fecal-oral route, and is therefore a key risk factor of waterborne diseases, especially in areas with low access to safe sanitation (Bridle, 2021; Prüss-Üstün et al., 2004). The economic history literature suggests that most of the decline in child mortality in high-income countries in the early 20th century can be attributed to investments in safe sanitation and drinking water (Alsan and Goldin, 2019; Cutler and Miller, 2005). Yet, recent RCTs on sanitation investments in low-income countries show mixed results (Schmidt, 2015) and almost one million children die each year from waterborne diseases (Prüss-Ustün et al., 2014; WHO, 2022a). Our second aim is thus to investigate whether preexisting sanitation mitigates waterborne diseases following an exogenous increase in stagnant water. We find that having access to improved sanitation attenuates most of the harmful effects of a stagnant water shock on child diarrhea and mortality.

We develop a novel method to predict stagnant water shocks based on established hydrological engineering principles. We use variation in rainfall, evaporation, topography, and soil infiltration from high-resolution satellite data to simulate surface water flow over 90 m grid cells across mainland Tanzania from 1999 to 2022. We focus on Tanzania because it is frequently afflicted by cholera outbreaks, is large enough to provide considerable variation in topography and climate, and offers

¹This statistic also includes food-borne illnesses, since the responsible pathogens spread through both contaminated water and food.

²For example, cholera causes acute diarrhea and is caused by the intake of cholera bacteria, and the Rota virus causes diarrhea and abdominal pain. This is in contrast to mosquito-borne malaria which is contracted in the bloodstream from a parasite and typically not classified as a waterborne disease.

detailed georeferenced data on both children’s health and learning from several large-scale surveys. The treatment variable, *Waterborne Disease Potential* (WDP), is defined as the share of the local area covered by stagnant water in the 8 weeks before a child is surveyed.

To the best of our knowledge, we are the first to produce a high-resolution, time-varying measure of stagnant water occurrence at this geographic and temporal scale, allowing us to match exposure to treatment to nationally representative survey data on children’s health and educational outcomes. To causally estimate the effect of WDP on children’s health and learning, we use repeated cross-sectional household survey data and a difference-in-differences identification strategy. For health outcomes, we use the *Demographic and Health Surveys*, specifically four waves of georeferenced data from 1999, 2010, 2015 and 2022, with which we estimate effects on short-run health symptoms, such as diarrhea and fever, and mortality for children aged 0-5 years. For learning outcomes, we use data from the Uwezo surveys, which provide standardized test scores and records of school absence for children, collected near annually from 2011 to 2017.

We find that a 10 percentage point increase in WDP in the two months prior to the children being surveyed leads to a 30% increase in diarrhea incidence, but find no effects on other health outcomes, suggesting that the negative impact of stagnant water is primarily due to waterborne diseases. Access to high-quality sanitation and improved water sources completely mitigates this effect. The effect of stagnant water shocks on diarrhea are larger in warmer and more densely populated locations, which poses challenges with ongoing trends in global warming and urbanisation. We also find that awareness of how to mitigate waterborne diseases does not increase with stagnant water exposure, neither in the short nor long run.

On learning, we find that exposure to a 10 percentage point increase in WDP in the two months prior to the children being tested lowers average test scores by 7% of a standard deviation and increases school absence by more than 5 percentage points. However, these results mask considerable heterogeneity, with much greater effect sizes in urban areas lacking adequate sanitation. The negative effect on learning is robust to heterogeneous and dynamic treatment effects ([de Chaisemartin and D’Haultfoeuille, 2022](#)), and is not explained by other confounding factors such as malaria, the disruptive effects of floods, or child labour.

Following these insights, we produce a back-of-the-envelope calculation of how the burden of waterborne diseases from stagnant water shocks is affected by global warming. Using the latest

climate projection data for East Africa ([Ayugi et al., 2021](#)), we estimate that 2°C of global warming may lead to a threefold increase in the combined severity and frequency of stagnant water shocks and consequently, waterborne disease outbreaks.

Our findings have a number of important policy implications. First, stagnant water is a preventable environmental hazard that can be prevented by improved sanitation and urban planning. Second, because of low awareness even in exposed areas, stagnant water shocks may be mitigated by short-range forecasts and information campaigns regarding their potential harm. In the short run, health consequences can be mitigated by targeted treatment post-infection such as oral rehydration salts, and in the long run, through targeted investments in water and sanitation.

This paper presents the first causal analysis, to our knowledge, of the effect of waterborne diseases on a national scale. Our treatment variable is specifically tailored to measure the risk of waterborne diseases. Most public health literature on waterborne diseases investigates the association between weather events—such as intense rainfall and high temperatures—and disease outbreaks ([Levy et al., 2016](#)). A large economics literature relies on exogenous variation in rainfall to proxy for income and health shocks in developing countries (e.g. [Maccini and Yang, 2009](#); [Shah and Steinberg, 2017](#)). However, rainfall may influence several channels that ultimately impact child health and learning ([Sarsons, 2015](#); [Mellon, 2022](#)). By using variation in stagnant water as our treatment variable and separately controlling for local rainfall, we disentangle the effects of stagnant water from the potentially confounding influence of local weather events.

Relatedly, this paper makes a significant contribution to the measurement of risk factors for waterborne diseases, with three specific advantages over existing methods. First, there is a general lack of stagnant water data with sufficient spatial resolution and frequency for large-scale panel data analysis. Satellite data tends to suffer from missing values due to increased cloud coverage during wet seasons and is primarily used for long-term occurrence analysis ([Pekel et al., 2016](#)).³ Meanwhile, available flood maps represent rare, spatially constrained events, while our measure captures the more frequent, day-to-day exposure to stagnant water. Moreover, since our prediction algorithm uses meteorological data, it can readily be plugged in to a short-range forecasting model to predict future disease outbreaks, used in climate change impact analysis, and applied to

³There is also a potential endogeneity issue in using observed surface water occurrence, since locations may differ in their ability to divert stagnant water in ways that correlate with our outcomes of interest. In contrast, the within-location variation of our measure is driven by arguably exogenous meteorological variation, primarily rainfall on upstream catchments, which permits a causal interpretation of our estimates.

additional countries or regions.

We contribute to a large economic literature on how shocks, in utero or early life, affect children’s human capital and economic outcomes. Existing studies show that children’s development is impacted by exposure to rainfall (or drought) shocks ([Maccini and Yang, 2009](#); [Shah and Steinberg, 2017](#)), floods ([Rosales-Rueda, 2018](#)), parasites ([Bleakley, 2007](#); [Miguel and Kremer, 2004](#)), and shocks in general ([Currie and Almond, 2011](#)). These studies tend to estimate shocks on a relatively wide early-life window (e.g. the year of birth) and find large long-run effects. We contribute to this strand of literature by investigating how exposure to stagnant water, at a well-specified time and within small spatial windows, affects contemporary health and learning outcomes. This paper is the first to causally address the effect of waterborne diseases on children’s human capital accumulation and at two key stages of child development: health shocks in young children, and the effect on cognition and absenteeism for school-aged children and adolescents.

Furthermore, we draw from a growing literature on how climate change will affect public health in the future and contribute to our knowledge on how extreme weather may propagate the spread of waterborne pathogens. Importantly, waterborne diseases are affected by the dynamic interaction of both temperature and rainfall. For example, [Levy et al. \(2018\)](#) in a recent review of the climate and waterborne diseases literature report a need to identify combined dynamic effects of temperature and precipitation. Indeed, the most recent IPCC report stated that climate change will increase the incidence of water-borne diseases with “high confidence”, and particularly in regions with limited safe water access ([International Panel of Climate Change, 2022](#)).⁴ We contribute to this literature first by quantifying how a warmer climate will lead to increased *severity* from stagnant water shocks, and then how a future increase in rainfall intensity will increase the *frequency* of these shocks, using the latest climate projections for East Africa ([Ayugi et al., 2021](#)) and future rainfall variability ([Li et al., 2021](#)).

Recent randomised controlled trials on sanitation investments have found small or no effects on both diarrheal disease and related mortality, sometimes due to low uptake rates ([Schmidt, 2015](#)).⁵ This is in stark contrast to historical data from the United States and Sweden, which shows that investment in safe water and sanitation reduced or eradicated mortality due to waterborne diseases

⁴See the IPCC 2022 Technical Summary B.4.5-6, page 50.

⁵But for an exception, see [Duflo et al. \(2015\)](#) who finds that sanitation investments, when integrated into a comprehensive water and sanitation program, led to a reduction in diarrheal episodes by 30-50%.

([Alsan and Goldin, 2019](#); [Cutler and Miller, 2005](#); [Knutsson, 2020](#)), and RCTs on water treatment which typically find high reductions in both diarrhea and child mortality ([Kremer et al., 2022](#)). We contribute to this literature by analysing to what extent access to improved water and sanitation mitigates the effects of stagnant water. Consistent with, [Duflo et al. \(2015\)](#) and [Alsan and Goldin \(2019\)](#), we find that water and sanitation investments have to be of sufficiently high quality to effectively mitigate the increase in diarrhea incidence.

Lastly, given the importance of cognitive skills for economic growth ([Hanushek and Woessmann, 2012](#)), this paper contributes to a broader literature on the geographic determinants of development by showing how geographically determined exposure to stagnant water affects children’s performance.⁶ Our contribution lies in introducing a novel measure of waterborne diseases risk and showing how this affects children’s human capital today at the micro level, which in turn may explain some of the subnational variation in development in low-income countries. In addition, the consistently stronger negative effects in urban settings suggest that waterborne diseases may not improve with economic development, but may instead be exacerbated if economic development leads to uncontrolled urbanisation without complementary investments in sanitation.

The rest of this paper is organised as follows: [Section II.](#) provides background on the role of stagnant water for waterborne diseases and the current health burden of these diseases in developing countries. [Section III.](#) summarises our data sources and key variables, while [Section IV.](#) explains our empirical strategy. [Section V.](#) presents our main results, [Section VI.](#) provides a battery of robustness checks and explores alternative explanations to our results. Finally, [Section VII.](#) provides a climate change impact analysis while [Section VIII.](#) concludes the discussion and presents our policy recommendations.

⁶Some of the most notable contributions are [Gallup and Sachs \(2000\)](#) who estimate the economic burden of malaria, [Easterly and Levine \(2003\)](#) who investigate the role of exposure to germs on contemporary development, [Nunn and Puga \(2012\)](#) who focus on the role of ruggedness on income today, and [Alsan \(2015\)](#) who estimate the effect of Tsetse fly on contemporary development.

II. Background

II.A. Waterborne diseases and stagnant water

Waterborne diseases are adverse health conditions caused by pathogens that are transmitted primarily through water or food. These pathogens include bacteria, viruses and worms, and common diseases are cholera, typhoid fever, and dysentery.⁷ Symptoms vary depending on the cause, but the most common by far are diarrhea and other gastrointestinal issues, such as abdominal pain (Magana-Arachchi and Wanigatunge, 2020). For most types of waterborne diseases, the incubation period is a few days to a few weeks (Percival et al., 2014), and disease outbreaks tend to occur in a matter of weeks after the first infection. For example, cholera takes between 2 hours to 5 days for a person to show symptoms after first ingesting contaminated food or water (Azman et al., 2199).

Waterborne pathogens occur naturally in aquatic environments, but tend to multiply under favorable conditions, such as in stagnant, warm, and polluted water (Bridle, 2021). Stagnant water, in contrast to flowing water, enables the formation of biofilms, the clustering of waterborne pathogens into communities that enable exponential growth and survival (Wingender and Flemming, 2011), which can develop in a matter of hours or days (Ling et al., 2018).

Stagnant water becomes an especially important transmission link in crowded environments with poor access to sanitation, such as urban slums. For most waterborne diseases, the fecal-oral channel, where contaminated human feces spread to water or food, is the most important channel behind outbreaks (Magana-Arachchi and Wanigatunge, 2020). In this process, waterborne pathogens can survive from ingestion to excretion and, when sanitation is inadequate, enter new water sources or agricultural environments. The formation of stagnant water through surface flooding creates a direct link between unsanitary conditions and human exposure (Prüss-Üstün et al., 2004).⁸ As such, the spread of waterborne diseases is closely tied to Water, Sanitation, and Hygiene (WASH) practices, and improving these practices can help disrupt the fecal-oral transmission pathway.

Most empirical work on the role of climatic drivers of waterborne diseases is in the form of time

⁷See Table B.1 for a summary of some of the most commonly occurring diseases and symptoms, by type of pathogen.

⁸See the diagram in Figure B.1 for a complete overview of transmission mechanisms of waterborne diseases through the fecal-oral channel.

series analyses and case reports, and typically finds seasonal correlations between heavy rainfall and high temperatures and local disease outbreaks (Levy et al., 2016). Effect sizes tend to be larger in warm areas, as warmth is often necessary to induce pathogen growth, and areas that have been dry, as this leads to a buildup of pollutants and fecal matter (Kraay et al., 2020). However, few studies have used variation in exposure to stagnant water, our main hypothesised transmission mechanism, explicitly. A few case studies have relied on variation in river levels as a proxy for stagnant water and analysed impacts on diarrhoeal incidence in individual villages or cities (Alexander et al., 2018; Hashizume et al., 2008), and typically find that this is a more informative measure than rainfall alone. A recent working paper analysed the effect of rainfall on a cholera outbreak in Dar-Es Salaam in 2015 and found that most of the effect from rainfall was driven by locations most vulnerable to flooding (Picarelli et al., 2017).

II.B. The waterborne diseases burden in Tanzania

Waterborne diseases rank as the second to fourth most debilitating disease in low-income countries depending on the measure used, but typically only represent a minor share of the disease burden in high-income countries. This difference is believed to be partly due to low access to safe sanitation and water, and it is estimated that 829,000 people die from a lack of safely managed drinking water each year (UN, 2022).

Despite this, low-income countries such as Tanzania has made great strides in reducing childhood diarrhoea mortality, with the mortality rate declining by 89% from 1980 to 2015 (Masanja et al., 2019). This is mostly attributed to the introduction of oral rehydration salts (ORS), while only 3% of the reduction is attributed to improved access to safe water and sanitation. The consequences of this is that waterborne diseases remain as prevalent as ever, as seen in Figure B.2. This is largely because ORS, in contrast to safe water and sanitation, do not prevent waterborne diseases but only mitigate the worst consequences. Repeated exposure to diarrheal episodes may affect gastrointestinal functioning, and increase the risk of malnutrition and stunting. Moreover, exposure to diarrhoeal diseases in early life is associated with lower cognitive ability in later life, even independent of stunting (Niehaus et al., 2002; Pinkerton et al., 2016). Hence, there is a great urgency to address not only the consequences of waterborne diseases on mortality but also their repeated occurrence.

While waterborne diseases are particularly disruptive for children under five, older children are also at risk. Prevalence varies smoothly with age, with a minimum occurring typically around the age of 15-20 years, as [Figure B.3](#) shows. Thus, we would also expect school-going children in the ages of 6 and above to also be affected by stagnant water shocks increasing the risk of contracting waterborne diseases.

In addition, there is large geographic variation in the waterborne disease burden also within low-income countries, which can partly be explained by the fact that local climate and geography play an important role for waterborne disease outbreak. [Reiner et al. \(2020\)](#) find that high-prevalence areas in Tanzania carry a three times greater burden than low-prevalence areas. In this regard, Tanzania is representative of many other low-income countries that face a continued persistence in waterborne diseases and an uneven distribution within the country.

For more context on waterborne disease burden in Tanzania specifically, see [Section C.1](#) and for a recent history on prevention and treatment strategies in Tanzania, see [Section C.2](#).

III. Data

III.A. Geographic data

To generate our treatment variable, WDP, we construct a high-resolution hydrological model to simulate the flow of surface water across all of Tanzania. Our aim is to create a treatment variable with the highest possible geographic resolution that is feasible enough for large-scale simulation, using harmonized high-resolution data as input to a hydrological model. To this end, we use the latest available global gridded datasets at the highest possible resolution, as shown in [Figure I](#). For the topographic data, which determines the flow of surface water between grid cells, we use data from the Shuttle Radar Topography Mission ([Farr et al., 2007](#)), representing the topography in Tanzania in year 2001, before our survey data begins. Rainfall and evaporation data, which allows water to enter and leave the model, are extracted from ERA5 reanalysis data. ERA5 is a dataset produced by the European Centre for Medium-Range Weather Forecasts using a combination of weather station and satellite data, coupled to a physical weather forecasting model, to generate a consistent, high-resolution long-term record of past weather and climate conditions ([Hersbach et al., 2018](#)). One advantage this has over historical data from rainfall measurement stations is

that it is less sensitive to measurement error stemming from a lack of rainfall station infrastructure, which might be correlated with local economic development. Finally, to account for soil infiltration, we use the ISRIC 2.0 global soil database (Poggio et al., 2021), which contains high-resolution data on soil composition across the globe. Section A.1 contains detailed descriptions of each of these data sources, while Section IV.A. describes the methodology we use to simulate stagnant water across Tanzania over time.

[Figure I about here.]

In order to investigate treatment heterogeneity in terms of local climate, we use historical annual mean precipitation and temperature from the WorldClim v2 database (Fick and Hijmans, 2017), covering the period 1970-2000.

III.B. Demographic and Health Surveys (DHS)

To explore health-related outcomes we use the Demographic and Health Surveys (DHS)⁹, specifically the four waves with georeferenced data: 1999, 2010, 2015, and 2022 compiled by IPUMS (Boyle et al., 2022).

DHS are repeated cross-sectional surveys representative at the national level, with two important features for our analysis. First, DHS provides spatial coordinates of villages (so-called ‘clusters’), although for confidentiality reasons the coordinates are randomly and uniformly shifted 0-5 km for rural areas and 0-2 km for urban areas. Since village locations will differ across waves, while we want to exploit the within-location variation in WDP over time, we assign each household to a 50x50 km static grid cell. The cells are large enough to capture the variation in the share of stagnant water over time, but arguably small enough to control for most of the unobserved geographical variation.¹⁰

The second advantage of DHS data is that it reports on health issues that children may have had in the last *two weeks* leading up to the survey date. We can therefore accurately capture short-term changes in health. DHS collects information on children’s recent health issues (for example, “has

⁹The DHS Program is funded by USAID and has collected survey data with a particular focus on children and women’s health since 1984. For more information, visit <https://dhsprogram.com/>

¹⁰This resolution also seems small enough to capture most of the subnational variation in diarrheal diseases, as depicted in Figure B.4.

your child had diarrhea in the past two weeks?”) as well as basic anthropometric features such as height for age and weight for age. Importantly, since the children surveyed in DHS are below five years old, they are likely more sensitive to waterborne diseases than older children of school age (Khalil et al., 2018). Assuming that the risk of contracting waterborne disease decreases with age, we can interpret these results as upper-bound estimates for children of school-going age.

III.C. Uwezo surveys

To analyse learning outcomes we use the Uwezo surveys. These are collected by the NGO *Twazeza* and are large-scale assessments of school-age children in Kenya, Uganda and Tanzania. The surveys are administered to schools, communities and households, and are repeated cross-sections. We use the survey waves from the years 2011, 2013, 2014, 2015 and 2017, all of which are georeferenced and representative at the district level. In our paper, we employ the household surveys for children’s test scores, and school surveys for absence.¹¹ For each household, there is information on household wealth and assets, as well as basic socio-economic information such as the child’s age, year of schooling, whether they are currently enrolled, mother’s age and mother’s schooling.

There are three features of the Uwezo surveys that are crucial for our purpose of studying the effect of stagnant water on learning. First, all school-aged children (aged 6-16) in the surveyed household are tested in basic Mathematics, English and Kiswahili by being administered a random sample of question cards in each subject. This allows us to measure downstream effects on learning and cognitive ability, in addition to first stage effects on health. Another advantage is that test scores are collected for all children of school-age in the household which makes us able to capture the effects also on those children who drop out of school due to e.g. ill health. Lastly, the surveys are administered by Twazeza. The NGO works largely independently of the Tanzanian government, and the tests have no bearing on the school grades of the children or the evaluation of teacher performance. Hence, there is little reason to believe that parents or schools would influence performance during the tests, which may be an issue when using test scores from school-administered tests.¹²

Depending on how well the child answers the question sets, the children get allocated a score:

¹¹For more information, visit <https://uwezo.tanzania.or.tz/>

¹²For a more extensive background on the education system in Tanzania and the current status of Tanzanian children’s learning, see Section C.3.

The scores are discrete and are 1-5 for reading in English and Kiswahili, and 1-6 for Mathematics, except in wave 2015 where scores are given between 1 and 9. For example, scores in reading from 1 to 5 are based on whether the child can recognise letters (1), recognise words, read a paragraph or read a short story (5)¹³. We have no reason to think that the effects would be different for verbal and quantitative abilities, and hence we construct our main outcome as the mean age-standardised test score across the three subjects. We standardise each child's test score by wave to have mean zero and standard deviation one, and then take the average and standardise again such that

$$\text{Mean standardised test scores} = \frac{\text{Avg Score} - \text{Mean}(\text{Avg Score})}{\text{SD}(\text{Avg Score})} \quad (1)$$

where *Avg Score* is the average for each child across the three subjects.¹⁴ We thus end up with a continuous measure of children's current performance and learning which is comparable across waves. In our results, unless otherwise stated we refer to the mean standardised test score as "test scores". Test scores are missing for 8% of our sample which we exclude from our analysis. If these differ systematically from the non-missing values this could bias our result. For example, if the test scores are missing because children did not get tested as they were ill, we may underestimate the effect of WDP on test scores.¹⁵

The second important feature of the Uwezo surveys is that we similar to DHS know the exact survey date for the vast majority of observations. The third important feature is that the data is georeferenced at the *ward*-level instead of at the village-level, which means that we can use ward-level fixed effects directly, instead of imposing a grid cell structure as we do for the DHS data¹⁶.

¹³In Mathematics, scores are based on whether the children can count, recognise numbers, order numbers, add, subtract or multiply.

¹⁴That is, for each child and $\text{Subject} \in \{\text{English}, \text{Kiswahili}, \text{Maths}\}$ we standardise each score by wave:

$$\text{Subject}_{\text{std}} = \frac{\text{Score} - \text{Mean}(\text{Score})}{\text{SD}(\text{Score})}$$

where mean and standard deviation, SD, are wave-specific. We then take the mean of the standardised subject-specific variables and standardise, again by wave, to construct *Avg Score*:

$$\text{Avg Score} = \frac{\text{English}_{\text{std}} + \text{Kiswahili}_{\text{std}} + \text{Maths}_{\text{std}}}{3}$$

¹⁵Assuming that children are too sick to take the test would have performed relatively worse, excluding these children would attenuate our results to zero. We test if WDP predicts the degree of missing values in test scores (for each separate subject) with our main model specification but we find no systematic correlation.

¹⁶The average ward area is 253 square kilometres, and the wards are of similar size as e.g. UK parliamentary constituencies.

To summarise, for health outcomes using the Demographic and Health surveys we create 50^2 km grid cells that contain villages from different waves, while for the Uwezo surveys we have data at the level of *wards*, whose boundaries are fixed over time.

IV. Empirical strategy

IV.A. Modelling waterborne disease potential

We are interested in estimating the causal effect of unsafe water on children’s health and learning. Since disease incidence likely correlates with unobserved geographic factors, we require exogenous variation in exposure to unsafe water in order to recover a causal estimate. A simple approach would be to use variation in rainfall to proxy for this risk, as this in many contexts show a clear association with diarrheal disease outbreaks (Levy et al., 2016; Lo Iacono et al., 2017). However, this is problematic for two reasons. First, stagnant water is often the result of rainfall on *upstream* catchments rather than local rainfall, and indeed in Tanzania, there seems to be no association between local rainfall and cholera outbreaks (Traerup et al., 2011). Second, local rainfall can affect children’s health and learning through multiple pathways not necessarily related to waterborne diseases (Mellon, 2022).¹⁷ To improve upon the rainfall measure, we use well-established hydrological modelling principles and high-resolution geospatial data to construct a nationwide hydrological model, which we then use to simulate the overland flow of water and the formation of stagnant water pools. Since it is the stagnant water that enables the growth of waterborne pathogens (Eisenberg et al., 2013; Leclerc et al., 2002), and we want to use variation in this measure as our treatment, we remove any permanent water bodies and free-flowing water from the simulation output before generating our treatment variable.

As input to the model, we use the global gridded geographic datasets described in Section III.A. and Section A.1. Simulations are run at a spatial resolution of 90 m, which is orders of magnitude smaller than the ward level but feasible enough to simulate over several years across Tanzania.¹⁸

¹⁷Stagnant water, which is typically water left standing after a flood, can also affect children’s outcomes in more ways than through an increase in waterborne diseases, which we discuss in Section VI.B..

¹⁸The runtime of hydrodynamic simulations are very sensitive to the level of resolution and typically with the *cube* of the spatial resolution. However, a recent evaluation of large-scale 2D hydrodynamic simulations for several European rivers show that resulting flooded areas and water levels are insensitive to variations in spatial resolution, once it is finer than 100 m (Dazzi et al., 2021; Falter et al., 2013, 2016). This is important as it allows us to feasibly simulate the formation of stagnant water across a whole country such as Tanzania over time.

For more details on the computational scheme, see [Section A.2](#).

We measure exposure to stagnant water (WDP) at the local level: for DHS we average over a circle with a radius of 10 km¹⁹ where the centroid is the village coordinate, and for Uwezo surveys we compute the average over each ward. We define WDP for each location and survey year, $S_{l,y} \in [0, 1]$, as the time average of the share of location l 's area, A_l , covered by stagnant water $A_{S,t}$ at day t over the past n days from the date of the visit:

$$S_{l,y} = \frac{\sum_{t=1}^n \frac{A_{S,t}}{A_l}}{n} \quad (2)$$

[Figure II](#), top row, shows the spatial distribution and resolution of the georeferenced locations in the DHS and Uwezo data. In the bottom row, we provide a three dimensional visualisation of the output from the hydrological model for an arbitrary point in time, as a first validation that our measure captures that stagnant water pools tend to form in low-lying areas prone to accumulating water. A more intense red colour denotes a higher value of $S_{l,y}$. Here we can see that stagnant water has been trapped in the lowlands next to the Lake Victoria coast, and in the valleys of a group of mountains.

In our main specification, we compute WDP as the time average over the last *eight weeks* prior to the survey date for each observation. We choose eight weeks mainly for two reasons. First, by limiting the analysis to within eight weeks, we hope to capture the short-run dynamics of disease outbreaks²⁰ and decrease the likelihood of our measure capturing other confounding factors, such as effects on agricultural productivity. Second, since we want to estimate the effects on children's learning and we do not necessarily want to capture the acute effects of illness, we look back over eight weeks to include children who got sick and then recovered²¹. In later robustness checks, we later relax this and find effects of similar magnitude and significance over the whole range of exposure over the last 1 to 12 weeks, which is the available extent of our data on stagnant water.

¹⁹The choice of radius length is partly to account for the fact that the village coordinate is shifted 0-5 km, and partly to create an area close to the average ward size.

²⁰Waterborne pathogens can grow and infect humans within days and even hours under ideal conditions, see [Section B.2](#) for more details.

²¹We think it is unlikely that children under acute illness were subjects to the test, even if they took the test in their home. We also do not find any association between WDP and missing data on test scores

IV.B. Validating Waterborne Disease Potential

To verify that our measure actually captures variation in surface water cross-sectionally and over time, we perform two validation exercises. First, using satellite data spanning 1989-2019 we find that WDP at the ward level is in accordance with satellite data, especially for non-permanent surface water, which is precisely what our treatment variable aims to capture. The model performs best for the treated locations²² which sees the largest dynamic variation in stagnant water, where our measure explains 18 to 39 per cent of the variation in satellite-measured surface water for transient water. This reassures us that our measure accurately captures the spatial variation in exposure to stagnant water across Tanzania.

To validate that our treatment also captures the *dynamic* variation within each location we rely on reported data at the household level from DHS. Conditional on location fixed effects, we find that exposure to WDP reduces the time to the nearest source of surface water, such that a fully treated location, with 100% stagnant water coverage, would imply approximately zero travel time to the nearest source of surface water. See [Section A.3](#) for the cross-sectional validation using satellite data and [Section A.4](#) for the dynamic validation using DHS data, as well as for a detailed methodology of the validation procedure.

[[Figure II](#) about here.]

IV.C. DiD model specification

We estimate the following difference-in-differences model for outcome Y of individual i residing in location l , surveyed in calendar month m in the year y :

$$Y_{ilmy} = \alpha_l^1 + \alpha_y^2 + \alpha_m^3 + \delta S_{ly} + \gamma R_{ay} + X'_{ily}\beta + \varepsilon_{ilmy} \quad (3)$$

where δ is the main coefficient of interest: $S_{ly} \sim (0, 1)$ is our WDP measure in each ward in the eight weeks prior to the date of the survey. Hence, δ captures the causal effect of an individual i being exposed to a higher likelihood of waterborne diseases in their location l in year y . Since the WDP share always lies between 0 and 1, the value of the coefficient presented in results below is the treatment effect when the stagnant water share is 100% of the local area – an unrealistic scenario.

²²Defined as those who switch in and out of treatment when using a binary indicator.

Thus, for interpretation of the results we refer to the coefficient divided by ten, reflecting a 10pp increase in WDP²³.

Next, as the share of stagnant water over time is potentially affected by local rainfall, and local rainfall may also affect the outcome, we include the last eight weeks of rainfall at the local area level for each wave, R_{ly} , in our baseline specification. We control for time-invariant unobserved differences across wards with ward fixed effects, α_l^1 , as well as survey-year fixed effects α_y^2 (equivalent to survey wave) and calendar month fixed effects α_m^3 to absorb seasonality effects, which have been shown to be informative of waterborne diseases (Reiner et al., 2020).

Lastly, we also include a vector of controls in X on the individual level (gender, age, whether the mother has secondary education or higher, the mother’s age, and an index for household wealth). We impute missing values within these variables as the sample mean.

To explore potential heterogeneities we also interact the variable of interest, S_{ly} with indicators for e.g. sanitation quality, an urban-rural indicator and other relevant heterogeneity dimensions. Formally, we estimate the triple DiD:

$$Y_{ilmy} = \alpha_w + \alpha_y + \alpha_m^3 + \delta_1 S_{ly} + \delta_2 WASH_{ily} + \delta_3 S_{ly} \times WASH_{ily} + \gamma R_{ly} + X'_{ily} \beta + \varepsilon_{ilmy} \quad (4)$$

where WASH is a binary indicator of sanitation status, such as whether the household has a toilet. Here, we are interested in δ_1 , δ_2 and δ_3 . The parameter δ_1 is the effect of WDP on our outcome when WASH=0, δ_2 is the direct effect on Y_{iwy} from WASH. The key contribution of this model is δ_3 , which is the coefficient on the interaction term. This estimate will give us the contribution to the outcome of the WASH variable through the effect of WDP. We will carefully discuss potential sources of endogeneity in this interaction variable, but we nonetheless believe these results provide important, if sometimes only causally suggestive, evidence.

IV.D. Descriptive statistics

Table I provides summary statistics of the DHS and Uwezo surveys, providing information on the individual and household level in Panel A and for the local area level in Panel B. For DHS, we have

²³While this is relatively uncommon, it is far from extreme. In the DHS data, we find this to occur in 5% of the sample, which is a far greater frequency than that of a devastating flood or other natural disasters.

information on 253 grid cells and on average 420 clusters (villages or neighbourhoods) per wave. There is large variation in household wealth and sanitation, and 78% of clusters are rural. Overall, WDP has a strong mode at zero, where the average level is 1.7%, but there is large variation (see also [Figure D.1](#) in [Section D](#) for a graphical representation of the distribution).

For the Uwezo surveys in the bottom half of [Table I](#), we include our main outcome, standardised test scores. The test scores are standardised with mean zero and standard deviation one by wave, thus the total sample has a slightly lower mean. Children are on average 11.1 years old, and 46% are girls. Households are relatively large with seven people on average. Mothers have a similar education to other national surveys (see [Section C.3](#)) where 24% have at least a secondary education. We also provide a wealth index based on normalised principal components of household assets, which is normalised to have mean zero and standard deviation one, by each wave. The components in the wealth index are the type of wall, whether the household has a radio, television, bicycle, motorbike, cattle, or electricity. Next, we see that 73% of households have any type of toilet. In panel B we provide ward-level characteristics on the 3876 wards. 84% of the wards are rural. In our sample, a ward includes approximately 20 households sampled per ward, but there is a large variation in both the number of households and villages sampled.

[[Table I](#) about here.]

[Figure III](#) shows the spatial distribution of average WDP across Tanzania at the ward level. While this is only representative of the time periods covering the Uwezo surveys, the distribution of treatment intensity compares favourably to the historical spatial distribution of cholera incidence, which shows a higher incidence in the Lake Victoria region, south-east, and north-west part of the country, where we also find greater likelihood of stagnant water forming ([Nkoko et al., 2011](#)). The spatial distribution of stagnant water in [Figure III](#) also compares favorably to the distribution of modelled diarrhea incidence using Bayesian geostatistical methods ([Reiner et al., 2020](#)), see [Figure B.4](#) for a map of this data.

[[Figure III](#) about here.]

One of the advantages of our methodology is to disentangle the effect of stagnant water from the effect of local precipitation. This is important, since rainfall can affect both income and child

labor demand through an increase in agricultural productivity, and may thus have an ambiguous net effect on children’s human capital through competing channels such as school absence and improved nutritional status (Shah and Steinberg, 2017; Maccini and Yang, 2009; Rosales-Rueda, 2018). In our main estimation we define and include a variable *Local Rainfall* which is equal to the sum (in m) of rain in the eight weeks preceding the survey date.²⁴ However, local precipitation may drive the estimated effect nonetheless. Column 1 of Table E.1 shows the correlation between the local rainfall and WDP. Although the correlation is positive, we find no statistically significant relationship between local rainfall in the past eight weeks and our stagnant water measure. This likely indicates that rainfall that contributes to stagnant water stems mostly from upstream catchments, and is consistent with the null finding on the relationship between rainfall and cholera in Tanzania (Traerup et al., 2011).²⁵ As the additional columns show, adding rainfall as a control variable does not impact the estimate in a meaningful way. While we are reassured that our results are driven by WDP and not local rainfall, we include it as a control variable throughout our analyses.

V. Results

V.A. Effects of waterborne disease potential on health

In the first part of our analysis, we examine the impact of WDP exposure on child health in Tanzania, using all available georeferenced survey waves from the DHS (1999, 2010, 2015, and 2022).

We first explore the effect of WDP on outcomes related to waterborne disease incidence. While DHS does not test for waterborne diseases specifically, they measure important physical attributes of children, such as weight for height, and recently experienced symptoms, such as diarrhea in the past two weeks. We use these variables as outcomes to test our main hypothesis that WDP increases waterborne disease incidence. We also conduct placebo checks on the effect of WDP on other health outcomes that are arguably unrelated to short-term stagnant water shocks, such as stunting.

²⁴We use eight weeks to mirror the time window used to construct the treatment variable.

²⁵Interestingly Traerup et al. (2011) finds that almost all cholera outbreaks tend to occur during the rainy season, but conditional on seasonal indicators, rainfall shows no significant relationship with cholera. This is consistent with a setting where local stagnant water will depend on upstream rainfall, and hence show a seasonal pattern, while being mostly independent of local rainfall, controlling for the seasonal effects which correlate with both upstream and local rainfall.

In [Table II](#), Panel A, we estimate our main DiD specification. The outcomes in the first two columns represent health states most plausibly affected by waterborne diseases: diarrhea, which is the symptom most closely associated with waterborne disease, and weight for height (since severe diarrhea tends to affect weight). Columns 3-6 represent outcomes that should arguably be less affected by a stagnant water shock: Fever, hemoglobin count, and height. The effect on fever might be positive if our results are driven by an increase in vectorborne diseases such as malaria, since fever is typically the primary symptom of these diseases. However, waterborne diseases may also cause fever as a secondary symptom. For that reason we separate between fever and fever without concurrent diarrhea. Hemoglobin count indicates whether the child has anemia. Anemia and height are arguably less sensitive to short-term stagnant water shocks. All health outcomes are equal to one if the child has had the health issue at any point in the past two weeks, while weight-for-height, hemoglobin count and height are tested at the date of the survey.

[[Table II](#) about here.]

We find a large and statistically significant effect on the probability that the child has had diarrhea: A ten per cent WDP increases the probability of the child having had diarrhea recently by 2.9 pp. Alternatively, a one standard deviation increase in WDP increases the probability that a child has had diarrhea in the past two weeks by 1.2 percentage points, which is an increase of 11% relative to mean incidence. The coefficient on weight is negative as expected but not statistically significant. For other and long-run outcomes, we find a significant and positive effect on fever, but not for those who have fever without concurrent diarrhea. Reassuringly, there is no effect of WDP on measured hemoglobin count or stunting, as measured by height, which could be the case if households select negatively into areas affected by short-term stagnant water shocks.²⁶

Next, we analyse the role of the child's water intake for the effect of stagnant water on diarrhea in Panel B of [Table II](#). In columns 1-2 we estimate the effect of WDP on diarrhea and split the sample by whether the child is breastfeeding. We hypothesise that children who are breastfeeding are less likely to have received food or water from other sources which should minimise the risk of contracting diarrhea from waterborne diseases. In these estimations we restrict the sample to

²⁶Interestingly, we find that the effect of WDP on anemia is positive and statistically significant but similar to the case for fever, we find no significant effect on anemia for children without diarrhea. We find no significant effect on cough, see [Table E.2](#).

children 24 months and younger, since breastfeeding at an older age is rare. We estimate that for children who do *not* currently breastfeed, a 10% increase in WDP increases the probability of the child having had diarrhea recently by 5.1pp (one sd of WDP increases diarrhea by 2.6pp). For children who breastfeed the coefficient is small and statistically insignificant. While breastfeeding is not randomised across children and these mothers or children likely differ in more than this aspect, the large difference in the effect of WDP is highly suggestive of a link between WDP, water and diarrhea, and consistent with public health advice to breastfeed to reduce diarrheal risk (Keskin et al., 2017; Bhandari et al., 2003).

In columns 3-4, we perform the same estimation of WDP on diarrhea but instead divide the sample by whether *the child has been given plain water in the past 24 hours*.²⁷ Consistent with WDP leading to an increase in the likelihood of water to contain pathogens, children who have been given water recently have a statistically significant 29pp higher likelihood of having had diarrhea recently. For the children who have not been given water recently, we find a statistically insignificant effect close to zero.

How households choose to hydrate their children during a stagnant water shock is likely endogenous, however, and may correlate with baseline child health. As a final placebo check, in columns 5-6 we again run the same specification and split the sample by whether the child was given water, but change the outcome to whether the child has had a fever recently. If the effect on fever looks similar, it might suggest a general issue with health in the community, or some interaction with the probability of contracting malaria. However, for both subsamples of children we find that the coefficient is small and not statistically different from zero. We take this as further support that WDP accurately measures risk factors in water becoming contaminated with waterborne pathogens.

1. Heterogeneity in effects on health

To better understand the effect we estimate, we next conduct a heterogeneity analysis both by key individual and location characteristics.

Gender: Existing literature shows that flooding affects children's outcomes differently for boys and girls (Maccini and Yang, 2009), and in India, the mortality rate for diarrheal diseases for girls

²⁷ As it is stated in DHS survey information.

have been approximately 50% higher than for boys (Morris et al., 2011). Similarly, as Table E.4 reports, we find a substantially larger effect of WDP on girls than for boys, with the effect size being more than twice as high for girls. However, the difference between the two is not statistically significant.

Age: Heterogeneity results for age are reported in Figure E.1. We find substantial age heterogeneity, where the effect for the age group 0-1 years is close to 0, and the highest effect is found for the age group 1-2 years. This is consistent with a ramping up of diarrhea infection rates when the child transitions out of breastfeeding (Bhandari et al., 2003). For age groups 2 up to 5 the effects are still positive, than smaller, and there is no sign that the effect attenuates with age, once the child is older than 2 years. This has important implications for our later results with test scores, as it suggests that exposure to stagnant water should have an impact also on children older than five.

Urban and rural: We next analyse if there are differences between urban and rural areas since most existing literature find that there are more severe disease effects in urban areas that are densely populated (Alsan and Goldin, 2019; Troeger et al., 2018). Table E.3 reports the results split by whether the surveyed location is urban or rural. We find that the effect is about 50% larger for urban areas²⁸. This suggest that there is a higher probability of contracting waterborne diseases in urban areas following a stagnant water shock, despite the fact that urban households tend to have better access to improved water and sanitation and are wealthier overall.

2. Heterogeneity by access to improved water and sanitation

Net, we analyse to what extent access to improved water and sanitation can mitigate the health consequences of a stagnant water shock. Existing literatures both in economics and public health emphasise the importance of the faecal-oral route in spreading waterborne diseases (Troeger et al., 2018; Magana-Arachchi and Wanigatunge, 2020), and in concordance with these, we find that access to high-quality sanitation completely mitigates the effects of stagnant water.

We classify household toilet types reported in the DHS into four categories: No sanitation, Unimproved sanitation, Shared sanitation, and Improved sanitation, based on the UN Sanitation ladder classification (Kvarnström et al., 2011).²⁹ We estimate the effect of WDP on diarrhea in one

²⁸Running the same specification with an interaction term yields a weakly significant difference between urban and rural locations.

²⁹To better understand the differences between these categories, Figure E.2 in Section E illustrates some examples.

estimation, interacting WDP with each category of the sanitation ladder, letting the lowest level of sanitation – no facilities and open defecation – represent the baseline. Of course, sanitation is not randomly assigned, but given the short-run nature of the stagnant water shock, we regard it as a predetermined variable unaffected by treatment variable in the short run. Given the existing evidence on the importance of the faecal-oral channel in spreading waterborne diseases, and our estimated results on the link between WDP and diarrhea, we expect a direct link between sanitation and waterborne diseases. Since both the existing literature and our previous results suggest that the WASH practices channel differ by urban and rural contexts, and given the different results in [Table E.3](#), we also split the sample by urban and rural location.

[[Figure IV](#) about here.]

The estimation results are summarised in [Figure IV](#). As expected, we see that WDP increases the probability of the child having had diarrhea recently for children living in both urban and rural areas, although the effect is both larger and more precisely estimated for urban areas. We estimate negative effects for all but one of the three interaction terms between WDP and each rung on the sanitation ladder. For the full sample, it is only households on the highest rung of the ladder where we can precisely estimate a mitigation effect of sanitation quality with regards to WDP. The coefficient suggests that households who have these sanitation facilities can completely offset the increased risk from WDP. This is consistent with existing evidence on the link between sanitation and diarrhea, which suggests that primarily *high-quality* facilities are important to stop the faecal-oral channel of the spread of waterborne diseases ([Troeger et al., 2018](#); [Magana-Arachchi and Wanigatunge, 2020](#)).

Furthermore, we see that the main negative effect of WDP and the mitigation with improved sanitation originate from urban areas, which is consistent with observational literature and historical case studies. The large heterogeneity between wards and households with higher and lower types of sanitation also provides an explanation for why only analysing the urban sample yields an imprecisely estimated effect, since there are large heterogeneities. For urban areas, we also see that the step from no facilities to 'unimproved sanitation' also contributes to mitigating the negative effect of WDP, but less so than the high-quality facilities. Interestingly, the effect of the third category, shared facilities, is estimated with the most imprecision. This may reflect that shared facilities

may be a poor indicator of better sanitation as there is large heterogeneity in how well they meet the sanitation needs of households (Magana-Arachchi and Wanigatunge, 2020). Indeed, we find that sharing a toilet with other households increases the risk of contracting WBD during a stagnant water shock, as reported in Table E.6.

Lastly, we interact WDP with a binary indicator for whether the household has access to an *improved source of water*, as defined by the WHO. The results are reported in Table E.5. We find that access to improved water dramatically attenuates the effect on diarrhea, but similar to the effects of high-quality sanitation, this matters only in urban areas. This implies that access to safe water is especially important in environments with a high presence of waterborne pathogens, such as densely populated areas afflicted by a stagnant water shock.

3. Effects on child mortality

Since diarrheal diseases are a leading cause of under-five mortality in Sub-Saharan Africa, and because of the significant effects of stagnant water exposure on diarrhea, we next examine if stagnant water impacts child mortality. We split the sample by urban-rural status and interact WDP with an indicator for improved sanitation, representing step 4 of the sanitation ladder, since this is where most of the mitigation of sanitation occurs.

Table III reports the effects of stagnant water on child mortality. Overall, there is a positive but insignificant effect of WDP on child mortality, and a negative but insignificant interaction effect. Similar to our results on the effect of stagnant water on diarrhea, however, this masks considerable heterogeneity between urban and rural areas. In urban areas, WDP has a statistically significant positive effect on child mortality over the past 8 weeks, and this is largely attenuated by having access to improved sanitation, although this interaction effect is insignificant. In contrast, there is much less attenuation from improved sanitation in rural areas. The results for the urban sample indicate that a 10 percentage point increase in stagnant water exposure doubles the baseline mortality rate. Moreover, the findings suggest that access to improved sanitation may substantially mitigate this effect, though the interaction term is not statistically significant. These results mirror our findings for diarrhea, and suggests that having access to improved sanitation prevent children from an increase in diarrhea, and may fully close the mortality gap as caused by stagnant water shocks. While access to improved sanitation itself is not randomised, it is arguably predetermined

at the time of stagnant water exposure.

The effect on mortality in urban areas is large and statistically significant. Additionally, since mortality is only tested related to the eight-week window since the household was surveyed, it is in combination with mortality as quite a rare event noteworthy that the estimates are relatively precisely estimated.³⁰ Assuming there is no stand-out effect in the two surveyed months, the total i.e. year-round, effect of stagnant water shocks on child mortality is likely higher than we can measure, causally. The mortality result also indicates that the effects we see on diarrhea and all later outcomes is on the survival set of children and thus also a lower-bound effect. Fortunately, mortality is relatively rare in our sample and in terms of the composition of surviving children should have a very marginal impact.

[Table III about here.]

V.B. Effects of waterborne disease potential on school performance

1. School absence

We next turn to educational outcomes, and assess whether the negative health effects from stagnant water shocks translate into worse school performance.³¹ We first analyse the effect on school absence. For this purpose, we use the school datasets of the Uwezo survey data, which contains a headcount of present children in the classrooms across different grades.³² From these data we construct our absence measure, a measure from 0 to 1 which indicates the share of enrolled children that are absent at the survey date. For sanitation, we lack detailed data on sanitation quality at the household level, and instead use the share of surveyed households that have access to a toilet within each village as a proxy for sanitation quality. This is a less endogeneous measure than household sanitation, and arguably a more accurate measure of the general sanitation level the child is exposed to given potential spillover effects of local sanitation practices.³³

³⁰As a placebo test, we assess whether childhood mortality outside this window is also affected by the treatment, and find no significant effects. Results available upon request.

³¹As a reminder, the children in these results are 6-15 years old as compared to 0-4 in the previous results.

³²School datasets are collected in conjunction with surveyed villages, and like all surveys in this study they are cross-sectional. Across our waves we have 15,428 school observations.

³³Conversely, in the DHS data, while sanitation quality is higher, the number of observations in each cluster is lower, which makes a village-level sanitation measure challenging.

[Table IV](#) shows the results for absence, by whether the ward is urban or rural, and by village sanitation. We find an overall positive effect of WDP on school absence. A 10pp stagnant water shock leads to 5.5pp (statistically significant) increase in absence, or 1 in 18 children. However – familiar from the results on child health for the younger children – we find that the overall effect masks considerable heterogeneity. In urban areas where sanitation is non-existent, the effect is more than three times as large. While the interaction effect with sanitation is insignificant, it is large in magnitude, and full village sanitation mitigates more than half offsets the increase in absence in urban areas caused by WDP. In contrast, there is no significant effect of WDP on absence in rural areas, and close to a zero interaction effect of village sanitation. This findings are largely consistent with the findings for child health, and shows that older, school-going children are also affected by stagnant water shocks, although only in urban areas, where high population densities facilitate the spread of these diseases.

As a sanity-check, we can compare the effects of WDP on diarrhea. Here, we find an effect size of 0.03 for a 10% WDP shock, which is comparable in magnitude to the effect size for absence for school children at around 0.05. While we might expect a smaller effect on absence, since school-going children are older than the children in the DHS sample, not everyone affected by waterborne diseases develop symptoms of diarrhea.

[[Table IV](#) about here.]

2. Test scores

We next analyze the effect of WDP on learning as revealed by standardised test scores. As discussed in [Section IV.C.](#), in our main specification WDP is the average share of ward area covered by stagnant water in the eight weeks preceding the survey and test date. Test scores are standardised averages of the child’s score in English, Maths and Swahili. Following our previous reasoning, we analyse heterogeneity by urban and rural ward and by sanitation quality through village-level sanitation.

[[Table V](#) about here.]

[Table V](#) reports the estimates of the effect of WDP on test scores. Similar to the effect on absence, WDP exerts a negative influence on test scores – a 10pp increase in stagnant water reduces

test scores by about 0.07 standard deviations. This represents a 1.2 standard deviations drop in test scores for each child absent due to the stagnant water shock. However, we cannot here distinguish between whether the child performs worse because they have been absent from school, or they are ill or just recovering from illness when they take the test and thus performing worse than they would otherwise. Even if the cognitive effect is only temporary, recent literature show there are significant permanent effects on education and lifetime earnings even from short bouts of school absence³⁴ This suggests that even one-time or infrequent exposure to stagnant water shocks may have permanent negative consequences. In [Section VI](#), we further explore how sensitive our results are to the time dimension of the definition of WDP.

Similar to our previous results, we find suggestive evidence for larger effects in villages lacking sanitation, especially in urban areas, and suggestive effects of a mitigating role of sanitation in urban areas (p-value 0.101), though the difference with rural areas is insignificant.

3. Heterogeneity in effects on school performance

Gender: Since we find larger effects of WDP on diarrhea for girls than boys, the effects on test scores may also differ by gender. In [Table F.2](#) we report the estimates split by gender. We find a slightly larger and more significant effect on female school children, although the difference between the genders is not significant. We are unable to test heterogeneity by gender for absence, since this is reported at the class level.

Age: Whether one would expect heterogeneity by age is ambiguous ex-ante. On one hand, younger children are arguably more vulnerable to stagnant water shocks and diseases. On the other hand, older children may receive less educational investment and be less likely to compensate for missing out on school due to sickness. For test score heterogeneity, we split the sample by age group, 6-10 and 6-11 years respectively, and report the results in [Table F.3](#). We find slightly larger and more significant effects on older children, though the differences in magnitude is small and insignificant. For absence, we split the sample by grades 1-3 and 4-7 and report the results in [Table F.4](#). We find an almost twice as high effect on absence for higher grades, where children on average are also older. This could possibly indicate that children in higher grades are more "marginal" students when it comes to enrollment, such that being struck by disease makes them

³⁴See e.g. [Cattan et al. \(2023\)](#).

less likely to attend school. Another way to interpret our findings is that the learning loss, measured by performance on standardised tests, per unit of absence loss is greater for younger children.

VI. Robustness Checks and Alternative Explanations

VI.A. Robustness checks

Non-linear treatment effects: In our main specification we estimate the effect of a continuous share of WDP (which can take any value between zero and one), which assumes a linear relationship between WDP and our outcomes of interest. To test robustness to this assumption, we bin WDP by magnitude. We create additional dummies of WDP and include these dummies in the same estimation (WDP 1-5%, 5-10%, 10-20%, and above 20%). These results are presented in [Figure E.5](#) for the effect on diarrhea, and [Figure F.1](#) for test scores. The effect increases monotonically with WDP magnitude for both outcomes, which would indicate support for a linear and continuous effect of WDP, although there are indicative signs of non-linearity at larger magnitudes. For diarrhea, we find no statistically significant effect of WDP for values less than 10%, while there are significant negative effects on test scores at values of WDP smaller than 10%. The estimates become slightly less imprecise with larger magnitudes, especially when $WDP \geq 20\%$, since these are relatively rare events (see [Figure D.1](#) for a histogram of WDP).

Randomisation inference: We conduct randomisation inference to evaluate to what extent our statistically significant findings could be based on spurious results ([Hsiang et al., 2014](#); [Young, 2019](#)). This is a powerful, data-driven method through which we can evaluate sensitivity to spatial correlation, non-random exposure to exogenous shocks ([Borusyak and Hull, 2022](#)), and the systematic assignment of placebo treatments in non-treated periods, commonly used in difference-in-differences specifications. We run our main specification from [Equation \(3\)](#), but randomly permute the treatment within each location, such that for every permutation, WDP values from a randomly selected year is assigned. The fact that for each iteration, all treatment values are drawn from the same year will take into account any spatial correlation between locations, and provide a data-driven way to estimate p-values that are adjusted for spatial correlation³⁵. The expectation is that

³⁵One advantage of randomisation inference over Conley standard errors adjusted for spatial correlation is that the former does not rely on assumptions about the correct distance cutoff values and functional forms of the spatial correlation structure.

we should find no effect on diarrhea or test scores with reshuffled treatment unless our results are driven by spurious regressions, i.e. the distribution of estimated coefficients should be centred around zero. We run these placebo regressions 500 times. [Figure E.3](#) and [Figure F.2](#) show the distributions of placebo coefficients for WDP for diarrhea and test scores. For both outcomes the effect of WDP is significant, based on the frequency of placebo coefficients that are greater than the estimated coefficient.

Heterogeneous and dynamic treatment effects: As we rely on variation in treatment timing, such that some locations are treated in different years than others, it is possible that our estimates for school performance are biased due to heterogeneous treatment effects ([Goodman-Bacon, 2021](#)). In our setting, the likeliest source of such heterogeneity is that children exposed to higher waterborne disease potential are sicker and more vulnerable, putting them on a permanently more negative trajectory compared to non-treated children.³⁶ If this trajectory persists and affects children across waves (i.e. years) a comparison against such wards would attenuate our effect to zero. Notably, since the vast majority of children in our sample are never treated, we do not expect such effects to dominate.

However to assess this issue, we implement the estimator developed by [de Chaisemartin and D'Haultfoeuille \(2018\)](#) (denoted DCDH). From the potential other estimators we implement this estimator since our setting is non-staggered, in that wards can go from treated to untreated and back ([de Chaisemartin and D'Haultfoeuille, 2022](#)). Moreover, since the correct estimators for continuous treatments where treatment is non-staggered are not well-established as of yet, we redefine our treatment, WDP, to a binary variable. This implies a ward becomes treated if the share of the stagnant water of the ward area exceeds 5%. [Table F.6](#) summarises these results. The estimate of the effect of WDP on test scores with two-way fixed effects (TWFE) is -0.085 standard deviations while the estimate with the DHDC estimator is -0.055, suggesting that heterogeneous treatment effects may play a role, although this estimate is also less precise, and the differences between the two estimates is insignificant. The DHDC estimator also allows us to study the effects of switching into versus out of treatment. If there is persistence in the effect over waves, a ward that switches out of treatment from one year to the next will arguably see a smaller effect than in wards where the control period is before the stagnant water shock occurred. This is precisely what we see.

³⁶This is in contrast to short-term health effects, such as diarrhea, where we are less worried about treatment heterogeneity and dynamics.

In wards that switch into treatment, going from below to above 5% stagnant water reduces test scores by 0.245 standard deviations, whereas the effect for those that switch out is close to zero and insignificant, although imprecisely estimated.

Parallel trends: Unfortunately, due to an unbalanced panel and a small number of waves, testing of pretrends yield mostly imprecise, although insignificant, estimates. As an alternative, in [Table F.1](#), column 5, we relax the parallel trends assumption, and allow wards with high average levels of stagnant water (those where the average level is 5% or more) to follow a different trend, by including the interaction with the continuous wave variable. This reduces precision slightly, but the estimate remains weakly significant at a similar magnitude. Alternatively, one can run a more demanding specification where every ward follows its own trend by interacting each ward with the wave variable. This results in a similar estimate as before, at the cost of less precision.

Sensitivity to choice of time window: In our main specification for test scores we define WDP to be the average share of stagnant water which covers the ward area over the past eight weeks. In [Figure F.3](#) we vary the number of weeks included in this average and re-estimate our main result, the effect of WDP on test scores. We find that the coefficient of the treatment is remarkably stable across the number of weeks included in the treatment window, and statistically significant throughout. To better understand which weeks drive this result, we next compute a treatment variable which measures the stagnant water share by week, discretely. That is, in one estimation we only include the stagnant water share in the third week since the date of the survey, and in the next, we only include the stagnant water share in the fourth week since the date of the survey. The estimation results are summarised in [Figure F.4](#) for test scores, and in [Figure E.4](#) for diarrhea. Here we see that there is a large initial effect which is measured with the most precision in week one, but the statistically significant effect remains up to eight weeks. This means there is likely a high serial correlation between these definitions of our treatment since the same pool of water can linger across weeks. Together, the results suggest that our estimated effect on test scores is the strongest for the weeks closer to the survey date but persists for the period observed. Importantly, we also see that the effect is measured also in the week immediately following the date of the survey, which is consistent with studies on outbreaks and the microbiological literature suggesting waterborne pathogens can contaminate and spread quickly.

Given that the results show high autocorrelation of WDP over time, one concern is that our

identification strategy captures the effects of a future stagnant water event. Such would be the case if high WDP in the recent past leads household to expect more favourable environmental conditions or events in the future in a way which affects test scores. If this is the case, *future* WDP (in terms of the date children are surveyed) should better predict the change in test scores than past events. However, due to the high correlation (95%) between weeks, models including weekly WDP in the same estimation are likely to induce problems associated with multicollinearity. [Figure F.5](#) displays two attempts to disentangle the potential role of future WDP while reducing the influence of multicollinearity: We measure the change in WDP over time or coarsely bin WDP across time. These results are measured with less precision than our main results, but past WDP better explains the reduction in test scores than future WDP (see [Section F.2](#) for more detailed explanation.)

VI.B. Alternative explanations

Malaria: Stagnant water may act as a breeding ground for mosquitos carrying the malaria-parasite as well as other vector-borne diseases, such as the West Nile virus and dengue fever. However, as [Table II](#) reports, we do not find a significant effect of stagnant water shocks on fever (column 4), the most common sign of malaria infection, unless diarrhea is also present (column 3). While this may be surprising, our identification strategy, which controls for location fixed effects, identifies the *short-term* effect of exposure to stagnant water (within 8 weeks of the survey date).

For mosquitoes to effectively spread malaria, they must first reproduce and their larvae undergo the full development phase before subsequently transmitting the disease to humans, which is estimated to take in the order of 8-12 weeks from the first rainfall to the first visible symptoms ([Teklehaimanot et al., 2004](#)). In contrast, waterborne diseases typically spread within 1-5 weeks, which is within the windows we use for the treatment definition. Secondly, there is typically a non-linear response between rainfall and malaria. Moderate amounts of rainfall tends to increase malaria risk, but too much rainfall may “wash out” mosquito larvae; empirically an inverted U-shaped curve between malaria risk and rainfall has been found in previous research ([Paaijmans et al., 2007](#); [Lowe et al., 2013](#)). For non-diarrhea fever, we observe the exact same relationship. [Figure E.6](#) shows that stagnant water over the past 8 weeks exerts a non-linear effect on non-diarrhea fever, which peaks at around 10% WDP and attenuates to zero at levels above 20% (though this is imprecisely estimated) . In contrast, the effect on diarrhea only increases with the magnitude of WDP, as seen

in [Figure E.5](#).

If malaria is less sensitive to short-term variation in stagnant water than waterborne diseases, one would expect a much stronger relationship between WDP and fever by leveraging cross-sectional variation. This is also what we observe. [Table E.9](#) reports the main health outcomes controlling for time, but not location, fixed effects. Locations with a higher baseline exposure to stagnant water reports much higher incidence of fever symptoms, including fever without diarrhea. We also find that in these locations, malaria awareness is very high, as reported in [Table E.10](#), Panel A, through measures such as whether malaria is reported as the most important health problem, and whether households know how to avoid and protect against malaria, through e.g. the use of bednets. Conversely, when we include location fixed effects in Panel B and instead leverage within-location variation in stagnant water over the short term, we do not find an additional increase in malaria awareness.

This suggests that areas more prone to stagnant water do indeed have a higher baseline risk of malaria, and that households are also informed of this risk, but, importantly for the interpretation of our main results, that malaria does not respond to short-term stagnant water shocks the same way that waterborne diseases do.

Consistent with our analysis of health outcomes, heterogeneity by baseline malaria prevalence suggests that the effect of WDP on test scores is not driven by malaria. We follow [Kuecken et al. \(2021\)](#) and use prevalence data of from the Malaria Atlas Project ([Sinka et al., 2012](#)) for 2010, the last year before our education data starts. [Table F.5](#) reports the results from our baseline specification run separately for high and low malaria prevalence areas, split by median malaria prevalence (approximately 7%). If anything, we find a substantially larger effect in areas of low malaria suitability, as measured by prevalence rates. While this heterogeneity may also indicate that potentially drier areas are less prepared against stagnant water shocks, the absence of a large effect of stagnant water in high-malaria areas suggests that malaria is not driving the effect on test scores.

Floods: For school performance, accumulation of stagnant water may confound damaging effects of floods on infrastructure, especially in rural areas, where children have to travel to school on unpaved roads. We believe that this is unlikely for several reasons. First, our exposure measure only takes into account *stagnant* water. This means that any flowing water is excluded from the treatment. Moreover, areas with sloping terrain, where flow velocities and thus erosion risk is much

higher, typically see little stagnant water. Second, we find significant effects at small doses of the treatment (around 5-10 %), while the effects seem to saturate at levels above 15-20%³⁷, suggesting that our results are not driven by large and disruptive floods. Third, our heterogeneity analysis suggests that most of the effects on test scores and school absence are driven by urban areas that lack access to improved sanitation, suggesting that the root cause is exposure to waterborne pathogens.

Agricultural productivity and child labour: One of the reasons why we control for local rainfall is to disentangle the results from the confounding effects of agricultural labour demand and income. It may however be the case that the accumulation of stagnant water is a useful resource for e.g. crop irrigation. Using crop calendar data for Tanzania, we do not find that the results change significantly for months where water availability is critical for plant growth.³⁸ Moreover, we find no significant differences between farming and non-farming households. If anything, as [Table V](#) shows, the treatment effects are larger for urban than rural households, suggesting that our results are not confounded by effects of local weather on child labour or income.

Anticipation and migration: We believe it is unlikely that households are able to anticipate stagnant water shocks, since there was no flood warning system in place during our period of study. Moreover, the short-term nature of the shock makes it unlikely that migration is biasing our estimates for health and learning.

VII. Impacts of and resilience to climate change

VII.A. The impact of climate change on waterborne diseases

How will climate change affect the frequency and severity of waterborne diseases in the future? The literature on climate and waterborne diseases posits that both temperature and rainfall are key determinants of waterborne diseases ([Levy et al., 2016](#)). High temperatures lead to faster potential pathogen growth, while intense rainfall events causing more stagnant water shocks are expected to become more common with climate change. Climate change is thus likely to affect both the *magnitude* and *frequency* of waterborne disease outbreaks. There is, however, still a lack of understanding on how the combined effects of these two phenomenon will affect waterborne diseases

³⁷Gradually excluding outliers with stagnant water values >50% down to >20% only results in larger and more significant treatment effects. Results available on request.

³⁸Results are available upon request.

(Levy et al., 2018). To estimate the combined effects of climate change on waterborne diseases, we use the latest climate projections for East Africa from Ayugi et al. (2021) in combination with our estimates from Section V.. A more detailed explanation of the underlying assumptions and results of related to the climate change analysis is provided in Section G.

The first-order effect of climate change is from warmer weather. In our main specification for diarrhea, we find a positive but insignificant interaction effect with temperature of 0.05 (p-value 0.36). With a baseline effect of WDP of 0.29, this suggests that each degree of warming may possibly increase the effect of stagnant water by 17 %. In Figure G.1 and Figure G.2 we do a more refined version of this analysis and split the sample by temperature windows of 2 and 4 degrees Celsius. This heterogeneity analysis similarly suggests that effects are greater in warmer areas, although the difference between warm and cold areas is not significant.

The second-order effect of climate change is through an increase in the frequency of intense rainfall events which ultimately drive stagnant water shocks. Using the latest CMIP6 projections, Li et al. (2021) find that for every degree of warming, the frequency of intense rainfall events in southern East Africa, where Tanzania is located, will increase by about 50%.³⁹

We can thus provide a first estimate of the impact of climate change by multiplying the magnitude and frequency effects, for each degree Celsius of projected warming. Figure V shows how much the total effect size increases relative to its baseline value for each degree C of projected warming. We can break this down into a “rainfall effect” and a “temperature effect”. Even assuming there is no interaction effect with temperature, the effects of climate change are substantial: the increase in intense rainfall means that 2 degrees of additional warming, relative to the study period, increases baseline risk by more than two-fold, and incorporating the temperature effects means that the increase is more than 3-fold. Because these effects are jointly driven by a warmer climate, the effects multiply and lead to an exponential-like increase in the risk of waterborne disease outbreaks.

While these estimates may seem large, they are comparable to recent epidemiological research, which has found that El Niño events, which are associated with both higher temperatures and more rainfall across East Africa, leads to a three-fold increase in cholera (Moore et al., 2017). These increases stem approximately in equal parts from direct effects of temperature and increased rainfall variability respectively, as Figure V shows. For more details on our climate change impact

³⁹This refers to rainfall events with return periods of 10-50 years, lasting 1-5 days, which roughly correspond to the return period of the large stagnant water shocks that drive our treatment effects.

analysis, see [Section G](#).

The range of considered temperatures is taken from two established climate change projections: SSP 4.5 and SSP 8.5 (Shared Socioeconomic Pathways).⁴⁰ SSP 4.5 indicates that the projected warming will be a “Middle of the Road” climate change scenario, whereas the SSP 8.5 scenario is likely an upper-bound, with dramatic temperature increases due to runaway increases in greenhouse gas emissions. This can be interpreted as the scenario where no improvements in climate change policy or emissions have been realised.

[[Figure V](#) about here.]

VII.B. Awareness and Preparedness for Waterborne Diseases

Given that climate change may imply a substantially greater risk of future waterborne disease outbreaks, it is crucial to understand to what extent past exposure to stagnant water exposure has facilitated adaptation through an increase in awareness and preparedness. Similar to our analysis where we explore malaria as an alternative channel of causation, we run an analysis both with-out and with location fixed effects, in order to gauge whether locations that have higher baseline exposure to stagnant water are more prepared for these shocks, and to what extent awareness is affected by a recent shock. We look at whether diarrhoea ranks as the most important community health problem, whether a mother is breastfeeding, whether a child was given plain water, whether a caretaker heard of or ever used oral rehydration salts (ORS), and where the household’s sanitation ranks on the sanitation ladder. Of course, we would not expect sanitation quality to change in the short term, but we include this for completeness.

Table [Table VI](#) reports our results. In Panel A, which relies on cross-sectional variation, we find that exposure to WDP has a *negative* effect on stated importance of diarrheal diseases. This negative effect is most likely explained by the fact that malaria is considered the most important problem in these areas, as discussed before in [Section VI.B.](#). When we include location fixed effects there is no significant effect. This suggests that malaria is perceived as a more salient risk in these areas, which could possibly crowd out efforts to prevent waterborne disease outbreaks. We find

⁴⁰See [International Panel of Climate Change \(2022\)](#) and [Section G](#) for more detailed descriptions. Note that SSP scenarios also take into account projections of technological advancement, population growth, energy policies and more.

that mothers breastfeed *less* in areas more prone to stagnant water, but again that there is no effect when including location fixed effects. This indicates that there is overall less breastfeeding in more vulnerable areas, and that breastfeeding does not increase during periods of increased stagnant water, despite its protective effects. Similarly, we find that giving plain water is not less common in areas more prone to stagnant water, and in fact *increases* during stagnant water shocks, which may be due to the fact that there is more surface water available for consumption. Reassuringly, there is greater awareness and use of ORS in locations with more stagnant water, but worryingly no increase in awareness or usage of this highly effective treatment following a stagnant water shock. This is consistent with evidence from Tanzania that suggests that local clinics are often lacking in crucial medical supplies such as ORS (Schellenberg et al., 2003), which makes them unable to respond to shocks in the short-term. Lastly, we do not find that sanitation quality is higher in areas more prone to stagnant water despite its mitigating effects. If anything, quality seems lower, though this difference is not significant. As expected, we also find no change in sanitation quality when including location fixed effects, as it is unrealistic that sanitation investments would be affected by a shock lasting only eight weeks.

[Table VI about here.]

VIII. Conclusion and Policy Implications

In this paper we have estimated the effect of exposure to stagnant water, a geographically determined risk factor of waterborne diseases, on children’s health and learning. We do this by constructing a hydrological model to simulate the local share of stagnant water, the WDP, in each location in the eight weeks prior to the date of collection of the health and learning data. We hypothesise that stagnant water pools encourage waterborne pathogens to proliferate: Greater exposure to this environmental hazard causes a local increase in waterborne diseases which affects both children’s health and capacity to learn.

Using a difference-in-differences identification strategy to estimate the causal effect of stagnant water exposure, we find that it primarily affects diarrhea – the main symptom of waterborne diseases – while having limited effects on other disease symptoms or long-run outcomes. Each 10 percentage point increase in the local area covered by stagnant water increases diarrhea incidence

by 30%.

Throughout our analysis, we find larger effects on diarrhea, mortality, test scores, and school absence in urban areas, and that the effect is largely mitigated by access to improved sanitation in particular. While sanitation access is not randomized, it is predetermined at the time of exposure, and its mitigation is consistent with the epidemiologically established fecal-oral pathway for waterborne diseases. Our results are consistent with historical events and documented disease outbreaks, where sanitation has been found to be a crucial factor in combatting waterborne diseases, especially in urban settings.

Policymakers in Tanzania, and other developing countries that battle recurring outbreaks of waterborne diseases, will likely have to invest in higher-quality sanitation facilities in order to reduce the incidence of waterborne diseases to that of the developed world. One way to make this more cost-efficient is to target areas more exposed to stagnant water.

While large-scale investment in WASH infrastructure is likely necessary, albeit expensive, in order to improve the local disease environment, an intermediate and potentially cost-efficient step in reducing the worst consequences of an outbreak could be improved targeting of medical treatment to areas at higher risk of outbreaks. We find that one standard deviation increase in WDP reduces test scores by 0.03 standard deviations. This effect size is equivalent to the effect on test scores estimated by [Mbiti et al. \(2019\)](#) who over a two-year period incentivise teachers in Tanzania with 5,000 TZS (3 USD) per student's passing grade (although their estimate is not statistically significant). The cost of this program can be compared to diarrhea rehydration treatment which costs only 0.56 USD. Despite this high cost-effectiveness, only about one in six children receive the treatment in rural Tanzania. We thus leave it to future research to explore demand and supply-side policies to increase children's access to treatment. Another lower-cost alternative to combatting the vulnerability for very young children is to campaign for increased breastfeeding, given the protective nature of breastmilk as indicated by previous research and by our results.

The burden of waterborne disease on both children's health and learning which we document in this paper implies that policies to combat waterborne diseases should also take future climate risks into account. Our projection of the future burden of waterborne disease shows a potential dramatic increase from the multiplicative effects of an increase in rare, intense rainfall events and warmer weather. Our results are especially concerning as access to water will be even more strained with

the onset of more severe climate change, as water shortage may increase the propensity and need of communities to draw water from unsafe sources. Combined with rapid urbanisation in developing countries, that may not keep up with necessary infrastructure investments, it is unfortunately likely that urban communities will see worsening public health status with regards to waterborne diseases. Our evidence of the mitigating effects of sanitation provides hopeful evidence on how the costs of waterborne diseases can be combated by targeted investment into vulnerable areas.

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Appendix for “Stagnant Water and Waterborne Diseases: Implications for Children’s Health and Education”

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A Geographic data and modelling stagnant water

A.1 Data sources

We use four different categories of data for our algorithm to compute WDP, as summarised in [Figure I](#) and [Table A.1](#), which are: Topographical data, rainfall data, evaporation data and soil infiltration data. See [Farr et al. \(2007\)](#); [Hersbach et al. \(2018\)](#); [Hersbach et al. \(2018\)](#); [Poggio et al. \(2021\)](#).

[[Table A.1](#) about here.]

We use topographical data from the Shuttle Radar Topography Mission ([Farr et al., 2007](#)). This is one of the most commonly used high-resolution global datasets on topography in scientific research, and is assembled by satellite data gathered in February 2000. Since we are not aiming to resolve detailed features in urban areas but rather large-scale runoff processes, we believe that this data is representative of the topography in Tanzania for the time period of our survey data. The resolution for each grid cell is approximately 30 m, with a vertical accuracy of around 6 m. Since we run our algorithm for the whole country, we aggregate this data to a 90 m resolution in order to make the computation feasible. This still leaves us with more than 120 million grid cells for the area of Tanzania and likely provides a detailed enough horizontal resolution for the large-scale analysis we run. Recent evaluations of large-scale 2D hydrodynamic simulations for several European rivers show that resulting flooded area and water level are insensitive to variations in spatial resolution once it is finer than 100 m ([Dazzi et al., 2021](#); [Falter et al., 2013, 2016](#)). The topography is used to determine the flow direction of each cell during the simulation, which may change dynamically as a function of the water depth over the cell, and to determine the water depth across cells at each point in time.

For rainfall input, we use data from ERA5, which provides a global gridded dataset with hourly estimates of a multitude of atmospheric variables, including precipitation ([Hersbach et al., 2018](#)). We use the reanalysis data, which is based on an ensemble of forecast models which take both satellite data and local weather station data into account and updates predicted atmospheric variables at a 1-hour resolution. The spatial resolution is 0.25 decimal degrees, which approximately translates to 27 km at the equator. This gives us close to 1400 data points for Tanzania, which

is vastly greater than the number of rainfall stations in the country. Gridded rainfall data derived from satellite observations is especially useful in developing countries, where there is often a relative scarcity and lower quality of data from rainfall gauges. Moreover, rainfall stations may vary systematically with local development, which means that measurement error may become systematically correlated with the local level of development. The advantage of using satellite-adjusted data is that measurement error is arguably orthogonal to local development. With regards to hydrological simulations, rainfall is especially important since it tends to be one of the main sources of uncertainty. Fortunately, a recent study investigating 22 global gridded rainfall datasets systematically found that ERA5 reanalysis data provided one of the best calibration scores and lowest inaccuracies when used operationally in hydrological models (Beck et al., 2017). The rainfall data provides all the input of water in the model, and thus drives the resulting surface runoff, depending on local infiltration rates, topography and evaporation.

For evaporation we also rely on the ERA5 reanalysis data, for the same reasons given above (Hersbach et al., 2018). This also lends consistency to the rainfall data, since these two datasets are produced jointly and dependent on each other. Specifically, we use the *potential* evapotranspiration rate which is applied to cells with a water depth greater than zero.

Lastly, for soil infiltration, we use soil data from the ISRIC 2.0 global soil database (Poggio et al., 2021). This is a state-of-the-art high resolution soil dataset which provides a resolution as fine as 1 km and provides a distribution of the content of clay, silt and sand at different soil depths. We depth-integrate this data and use the distribution of soils in each cell to classify each cell as a soil type according to the USDA classification system. From this classification we can then derive soil parameters such as saturated infiltration capacity and soil porosity. The current version of our algorithm uses a simplified infiltration measure by applying only the saturated conductivity of the soil to account for losses due to soil infiltration. Due to the high resolution of the topography we do not apply any slope-adjusted infiltration rates as some low-resolution hydrological models do. Instead a greater slope will translate into a faster runoff process, which will reduce the resulting infiltration. Future iterations of this algorithm could potentially be improved by applying a full soil infiltration mode commonly use in state-of-the-art hydrological models, such as the Green-Ampt method, and also by simulating the groundwater storage layer as a separate entity able to refeed the infiltrated water as groundwater seepage into rivers and streams, which as of current is not handled

by the algorithm.

A.2 Algorithm for WBD Potential

The purpose of the algorithm is to model the time evolution of stagnant water surfaces over time, which we wish to aggregate to a weekly-level treatment measure, using a combination of hydrological and hydraulic calculations. For this purpose, we run the model with 5-minute temporal resolution. Before the model is run, input data are processed to cover the same extent and transformed to the same coordinate system. Below is a schematic explaining how the algorithm, which is implemented in Python, works.

1. Initialise and run the model starting > 3 months before the first interview date until the last interview date.
 - (a) Initialise a new day d
 - (b) For every 5-minute timestep $t \in \{1, \dots, 288\}$ in each day:
 - i. Accumulate the last 5 minutes of precipitation on all grid cells
 - ii. For each cell i, j identify the immediately neighboring cell with the lowest current water level, where water level is the sum of the current water depth and elevation of the cell.
 - iii. If this cell has a lower water level than cell i, j , then transfer the 5-minute accumulated volumetric flow V from cell i, j to the receiving cell, otherwise do nothing. The volumetric flow rate $V = f(\Delta h)$ is an increasing function in the difference in energy levels (Δh) between the emitting and receiving cells, based on the Manning formula for open channel flow.
 - iv. Finally, remove the 5-minute accumulated volumetric infiltration rate from cell i, j : $\min(d_{i,j}, f(K))$ where $d_{i,j}$ is the water depth of cell i, j and K_s is a soil infiltration parameter. The current version, $f(K) = K_s$ where K_s is the saturated hydraulic conductivity of soil s in cell i, j .
 - (c) At the last time step of each day, $t = 288$, subtract the volumetric actual evaporation rate E from all cells where $E = \min(d_{i,j,t}, E_d)$ where $E_{i,j,d}$ is the potential evaporation rate in cell i, j for day d .

2. Export water depth for all cells to daily georeferenced arrays, which are then aggregated into weekly-level treatment measures at the ward-level using ArcGIS and the ArcPy package for Python.

A.3 Validation with satellite data

To validate that our measure of stagnant water, WDP, correlates with real-world observations of surface water, we use data from the Global Surface Water dataset, developed by the Joint Research Centre of the European Commission (Pekel et al., 2016). This dataset uses satellite imagery to identify surface water around the world at a 30 m resolution, spanning the time period 1984-2021 (Pekel et al., 2016). Each cell represents a value between 0 and 100, representing the per cent of monthly observations where the cell was identified as being covered by surface water. The raw data has a monthly resolution, but due to a large amount of missing data at this level, we compare the aggregated output from the model across all wards from the Uwezo dataset.⁴¹

We start by aggregating the Global Surface Water data at the ward-level, which gives us a measure of the average share of land covered by surface water over the period 1984-2021 within each ward (“All surface water”). We then do the same and include only cells where surface water has an occurrence rate between 0 and 50 per cent (“Intermittent surface water”) and 0 and 25 per cent (“Infrequent surface water”). This is because our stagnant water measure is supposed to capture variation in *impermanent* surface water, which drives the variation of the treatment. We remove any wards within 10 kilometers from the coast, to exclude the influence of variations in sea levels, which is unrelated to our measure of stagnant water.

Table A.2, column 1, reports the results of a regression of the average observed surface water on simulated stagnant water (WDP). The relationship is positive and strongly significant, although R^2 values are low, in the range of 0.02 to 0.07. However, what we care most about is to accurately predict stagnant water in those areas that drive the variation in the treatment in our sample. We define the wards that switch from below to above 5% and 10% “Switchers (5%)” and “Switchers (10%)”, respectively. For this subsample, the R^2 is considerably higher. For example, for infrequent surface water that only occurs 0-25% of the time, the R^2 for wards that switch into and out of the

⁴¹We could also do this validation for the DHS data, however, the ward data from Uwezo captures the full extent of Tanzania, and without any overlaps, and hence provides a more complete dataset.

binary treatment classifications range from 0.48 to 0.50. Note that we would not expect R^2 close to 1 since we are comparing long-term averages in the Global Surface Water dataset with averages computed over a few years for the WDP measure.

[[Table A.2](#) about here.]

A.4 Validation with DHS data

To complement the satellite validation and ensure that our measure captures the *dynamic* variation of surface water over time in each location, we use a variable reported in the DHS data: The amount of time (in minutes) the household has to their main source of water. We estimate the effect of WBD Potential on the time it takes to the water source, and also divide the sample by the source of water of the household: Whether they preliminarily obtain water from a tap, well or from a natural source of surface water. Intuitively, if WBD Potential accurately captures an increase in stagnant water, the amount of nearby surface water in the ward should increase, reducing the time it takes to get to the closest water source *only for households that collect water from nature*. [Table A.3](#) reports these results. We find that higher WBD potential leads to a lower reported time to nearest water source only for *natural* sources of water, such as lakes, ponds and rivers. Relating the coefficient to the mean of the dependent variable, we find that a 100% treated area, a fully flooded location, would lead to a close to zero distance to the nearest source of natural water. This suggests that an increase in WBD Potential leads to a measurable increase in the local stagnant water share reported by water-collecting households, and suggests that the model captures dynamic variation in stagnant water, in addition to the long-term cross-sectional variation reported in [Table A.2](#).

[[Table A.3](#) about here.]

B Detailed background on Waterborne Diseases

B.1 Mechanisms for contagion and disease symptoms

Waterborne diseases are adverse health conditions caused by pathogens that are transmitted by the intake of or contact with pathogen-polluted water, such as by the intake of harmful bacteria or worms. These pathogens include bacteria, viruses and worms, and common diseases are cholera,

typhoid fever, and dysentery. [Table B.1](#) summarises some of the most commonly occurring diseases and symptoms, by type of pathogen. Symptoms vary depending on its cause, but the most common one by far is diarrhoea and other issues relating to the gastrointestinal system, such as abdominal pain ([Magana-Arachchi and Wanigatunge, 2020](#)).

[[Table B.1](#) about here.]

For clarification, neither malaria nor chemically polluted water that causes health issues are typically considered waterborne diseases. First, malaria is not a waterborne disease. Malaria is a serious and sometimes fatal disease which infects humans via a parasite carried by mosquitoes (and not through contact with contaminated water per se) ([WHO, 2019](#)). Malaria causes flu-like symptoms such as fever and vomiting, but is in contrast to waterborne diseases not associated with diarrhoea. Second, chemically polluted water is not a waterborne disease. While water polluted with e.g. pesticides ([Boedeker et al., 2199](#)) or arsenic ([Mandal et al., 1996](#)) lead to large adverse health effects, they are not waterborne diseases since this term typically is reserved for pathogen-induced disease.

B.2 Spread, infection and recovery of waterborne diseases

Waterborne diseases spread via water, either by directly drinking water infected with waterborne disease pathogens or by swimming in contaminated water. For most waterborne diseases, the so-called faecal-oral channel where contaminated human feces spread to water or food which then is ingested by another human is the most common source of an outbreak ([Magana-Arachchi and Wanigatunge, 2020](#)). For example, a person could become infected by eating food prepared from agricultural output using human faeces as fertiliser. Thus, the spread of waterborne disease is inextricably associated with Water, Hygiene and Sanitation Practices (WASH) and improving these can break the faecal-oral transmission channel.

Waterborne pathogens occur naturally in rivers and lakes, but grow exponentially under conditions with stagnant water that become contaminated. Stagnant water enables pathogens to form biofilms and cluster close together which enables faster reproduction. Poor sanitation means an outbreak can spread across persons and households via the fecal-oral channel. In a lab-controlled environment, exponential growth of bacteria causing waterborne disease can occur within hours

(Farhat et al., 2018; Zlatanović et al., 2017). Ling et al. (2018) found that the whole water supply of a Chinese city became contaminated within six days under conditions where water stayed stagnant in the city's plumbing system.

For most contractions of waterborne diseases, the time from first contact to disease outbreak is a few days. For example, cholera takes between 2 hours to 5 days for a person to show symptoms after first ingesting contaminated food or water (Azman et al., 2199). Symptoms can last from days (e.g. a virus infection) to years (some types of worms), but most common is days or a few weeks (Percival et al., 2014). In severe cases waterborne disease can be fatal, in particular for young children, with severe diarrhoea leading to dehydration and death if left untreated.

[Figure B.1 about here.]

B.3 Waterborne disease burden in Sub-Saharan Africa

According to the UN, the Sustainable Development Goal to achieve safe drinking water for all will fall short of 1.6 billion people by 2030 (UN, 2022). The consequence of this shortfall is dire: The United Nations estimate that 829,000 people die from a lack of safely managed drinking water each year (UN, 2022). Strikingly, the burden of waterborne disease caused by unclean water fall almost exclusively on developing countries and in particular Sub-Saharan Africa (Anthonj et al., 2018). For instance, Black et al. (2010) estimate that each day 2,000 children under the age of five die in Africa due to diarrhoea – the second largest source of child mortality⁴².

However, the mortality number masks an even higher incidence of disease: Troeger et al. (2018) found that average diarrhoea episodes per person in Sub-Saharan Africa were 1.05 per year. Figure B.2 shows that despite improvements in child mortality due to the introduction of treatments such as ORS, incidence remains high and has barely changed in the last 30 years. Thus, waterborne disease does not only pose a fatal risk, but also when less severe is likely to affect the vast majority of people in the region several points in their lives.

In terms of age heterogeneity, the most vulnerable are typically young children and old individuals. Figure B.3 shows diarrhea prevalence in Tanzania by age group. There is typically a large reduction in prevalence from the age of below 1 years to 5 years, but after this it tends to flatten out.

⁴²18% of total mortality, the largest source of mortality being neonatal conditions in the first 28 days of life for infants (Black et al., 2010)

In the DHS data, we do find larger baseline diarrhea prevalence for 1-year olds which then reduces by age. However, as [Figure E.1](#) based on our empirical analysis shows, the *increase* in diarrhea risk following a stagnant water shock does not follow the same age pattern as baseline diarrhoea, but instead seems to peak at age 1-2 years, around the age where most children are no longer breastfed, and flattens out after this, with children up to 5 still seeing a large increase in diarrhoea prevalence from exposure to stagnant water.

[[Figure B.2](#) about here.]

[[Figure B.3](#) about here.]

B.4 Geographic distribution of diarrhoeal disease in Tanzania

[[Figure B.4](#) about here.]

C Institutional background

C.1 Prevalence and causes of waterborne disease in Tanzania

Waterborne diseases are illnesses caused by pathogens that are transmitted through contaminated water sources⁴³. Tanzania lies in the ‘belt’ of the world’s highest waterborne disease incidence, which stretches through central and eastern Sub-Saharan Africa, a region characterised both by a favourable climate for waterborne pathogens as well as lack of access to safe water and sanitation. The number of diarrheal episodes per person per year is 1.1 episodes in Eastern Sub-Sahara where Tanzania is located, which is only second in the world to Central Sub-Saharan Africa (1.21) ([Troeger et al., 2018](#)). Tanzania has had several outbreaks of waterborne diseases such as cholera in the past few decades. A recent example of a cholera outbreak occurred around Tanzania’s capital, Dar es Salaam, in 2015, with 16,521 reported cholera cases in total ([Chae et al., 2022](#)), but other waterborne disease outbreaks have been documented as recently as 2022 ([Masunga et al., 2022](#); [WHO, 2022b](#)). Waterborne diseases have historically been classified as ‘neglected’ diseases by Tanzanian authorities and have received much less attention than officially prioritised diseases such as HIV and malaria ([Tanzania Ministry of Health and Social Welfare, 2008](#)), despite the fact that

⁴³See [Section B](#) for more details on the definition and characteristics of waterborne diseases

waterborne diseases carry a higher disease burden than HIV and Malaria for children in Tanzania (Vos et al., 2020).

Environmental factors play a critical role in enabling the survival, growth and transmission of waterborne pathogens⁴⁴. El Niño events, which occur every 3-5 years, and lead to an increase in rainfall and flood events across East Africa are associated with a three-fold increase in cholera incidence (Moore et al., 2017). In a review of the epidemiological literature, Levy et al. (2016) found that the most important predictors of diarrheal disease outbreaks were floods, followed by heavy rainfall and high temperature. Heavy rainfall is thought to affect diarrhea incidence primarily through its effect on floods and surface water contamination (Levy et al., 2016). Detailed mapping in a district in Tanzania showed that low-lying areas with high water-tables are more susceptible to cholera as a result of rainfall, as this leads to the accumulation of surface water which contaminates pit latrines and shallow drinking water wells (Mayala et al., 2003), which lends support to our empirical strategy.

C.2 Prevention and treatment of waterborne diseases in Tanzania

Tanzania has made great strides in reducing childhood diarrhea mortality, with the mortality rate declining by 89% from 1980 to 2015 (Masanja et al., 2019), but incidence remains largely unchanged. Most of the lives saved are attributed to treatment with oral rehydration salts (ORS), which replenishes fluids lost by diarrhea. This makes children more resilient to the worst consequences of diarrhea, but does not affect the probability of contracting the disease in the first place. These are relatively affordable on the global market, costing close to 0.56 USD per treatment (UNICEF, 2022). In contrast, less than 3% of the reduction in diarrhea-related mortality is attributed to improvements in water and sanitation, which is evident by the fact that higher-quality sanitation coverage increased only from 8 to 15% over the same period (Masanja et al., 2019). Hence, the focus has been on treatment rather than prevention, which has left room for large sub-national variation in disease incidence (Reiner et al., 2020). In high-income countries, large-scale historical investments in water and sanitation have been crucial in reducing childhood mortality and diarrheal incidence (Alsan and Goldin, 2019). However, this infrastructure is costly, and lower-cost piecemeal approaches in Tanzania have been unable to curb diarrhea-related morbidity (Briceño

⁴⁴See Section B.2 for more details on transmission mechanisms

et al., 2017).

Treatment with ORS or antibiotics is usually administered at health clinics, which implies that access to high-quality healthcare is necessary to relieve consequences of waterborne disease infection. In 1984 Tanzania set up the National Control of Diarrheal Disease to combat childhood mortality, which focused on creating local clinics that could administer ORS (Masanja et al., 2019). This may explain why care-seeking for diarrhea in children in Tanzania is among the highest in Sub-Saharan Africa, with more than half of children sick with diarrhea taken to a health clinic (Schellenberg et al., 2003), and more than 90% live closer than 5 km from a primary health facility (Tanzania Ministry of Health and Social Welfare, 2008). However, this masks large inequalities in the quality of received healthcare. A study in rural Tanzania found that only 1 of 6 children with diarrhea received ORS, citing low diagnostic capabilities and a lack of medical supplies (Schellenberg et al., 2003), while other parts of Tanzania see ORS administration rates over 50% (Masanja et al., 2019). It is not uncommon for rural families to bypass the local health clinic, especially if living close to a hospital (Kahabuka et al., 2011), which suggests that while there is high access to health clinics, quality is often poor and unevenly distributed. Thus, we expect waterborne diseases to spread more easily in urban, densely populated areas, where contamination of water sources is more common, but that health consequences conditional on having contracted a waterborne disease may be worse in rural areas with lower access to high-quality healthcare.

C.3 Education and learning in Tanzania

In Tanzania, schooling starts with pre-primary schooling at ages 5-6, although it is common that children also attend pre-school for 2-3 years, which by global standards is relatively formal schooling (Bietenbeck et al., 2019). This is followed by seven years of primary schooling at ages 7-13, four years of ordinary secondary school (ages 14-17), and two years of advanced level secondary school (ages 18-19)⁴⁵. The adult literacy rate was 77.5% for men and 62.2% for women in 2012, but literacy among current pupils is higher, with the literacy rate being 86% for 15-24 year-olds. In 2012, 83% of the population reported having attained primary schooling, and 12.9% secondary schooling. Schooling expansion has been swift in Tanzania, and for example ordinary secondary school enrolment has increased from 6% in 2002 to 34% in 2013.

⁴⁵This information and subsequent on Tanzania school system from UNESCO (2014) and World Bank (2016)

The effect of waterborne diseases on learning and education as a whole is understudied, but there are key links between health and the state of learning in Tanzania. For one, many students drop out and do not finish primary schooling; 65.1% (72.8%) of boys (girls) finish primary schooling. Most dropouts are due to unknown reasons (truancy), but health reasons are common. For one, early-life stunting, from e.g. repeated exposure to diarrheal diseases, is an important cause of both absence and lower cognitive ability. Second, current health issues also lead to greater dropouts. For example, in 2014, a survey conducted in three Tanzanian regions stated that health problems caused 6.2% of dropouts, and in a 2013 school census, 3% of survey respondents reported dropping out due to illness or having to take care of someone ill ([UNESCO, 2014](#)). Even school absence as short as 10 days has been shown to have long-term consequences ([Cattan et al., 2023](#)) indicating that even if health is only temporarily affected, losses from learning may be permanent. Lastly, even when attending school, learning can be limited: In a survey performed by the World Bank, only 40% of students in year 4 (mainly ten-year-olds) could perform a year 3 mathematics task such as $6 \div 3$ or 7×8 . An important cause of limited learning is low quality of educational resources ([Mgema, 2022](#); [Ilori and Mlavi, 2016](#)), however fatigue or frequent absences due to waterborne diseases are likely to also negatively affect the capacity for learning.

D Descriptive statistics about the treatment

[[Figure D.1](#) about here.]

E Additional results with DHS

[[Table E.1](#) about here.]

[[Table E.2](#) about here.]

[[Table E.3](#) about here.]

[[Table E.4](#) about here.]

[[Figure E.1](#) about here.]

[[Figure E.2](#) about here.]

[[Table E.5](#) about here.]

[[Table E.6](#) about here.]

[[Table E.7](#) about here.]

[[Table E.8](#) about here.]

[[Figure E.3](#) about here.]

[[Figure E.4](#) about here.]

[[Figure E.5](#) about here.]

[[Figure E.6](#) about here.]

[[Table E.9](#) about here.]

[[Table E.10](#) about here.]

F Additional results with test scores

F.1 Results with test scores: Specification and heterogeneity

[[Table F.1](#) about here.]

[[Table F.2](#) about here.]

[[Table F.3](#) about here.]

[[Table F.4](#) about here.]

[[Table F.5](#) about here.]

[[Table F.6](#) about here.]

F.2 Results with test scores: Robustness

[Figure F.1 about here.]

[Figure F.2 about here.]

[Figure F.3 about here.]

[Figure F.4 about here.]

[Figure F.5 about here.]

Given that the results show high autocorrelation of WBD Potential over time, one concern is that our model captures a phenomenon in the future, which we then estimate effects for due to the correlation with past events. Such would be the case if past WBD Potential captures that children expect more favourable environmental conditions or events in the future in a way which affects test scores. If this is the case, *future* WBD Potential (in terms of the date children are surveyed) should better predict the change in test scores than past events. However, due to the high correlation (95%) between weeks, models including weekly WBD Potential in the same estimation are likely to induce problems associated with multicollinearity. Figure F.5 displays two attempts to disentangle the potential role of future WBD Potential while reducing the influence of multicollinearity. First, in the left subplot, we estimate a model with two variables: Our main WBD Potential measure (average eight weeks pre survey), and the difference between WBD Potential with that measure and the WBD Potential as measured two weeks after the test was taken. Thus, the coefficient would measure the marginal effect on test scores from the change in WBD Potential eight weeks prior to the survey and two weeks past. We find that the main measure has a negative effect on test scores consistent with our main result, while the "Future minus Past WBD Potential" is measured with a coefficient close to zero and more imprecision. The right subplot Divides up WBD Potential into three time periods across a longer time span: WBD Potential 2-9 weeks prior to the survey, WBD Potential one week on either side of the week of the survey, and WBD Potential measured 2-3 weeks after the survey. As expected from our hypothesis on the mechanism and the results in Figure F.4, we find that the WBD Potential 2-9 weeks prior to the survey is negative with a similar magnitude to that of our main results with the other two indicators centred closer to zero. However all three coefficients are statistically insignificant from zero.

G Impacts of climate change

G.1 Climate and waterborne diseases

Climate change is expected to lead both to greater variation in rainfall and increased temperatures, which may lead to large increases in the incidence and distribution of waterborne diseases due to multiplicative effects (Levy et al., 2018). A study by Traerup et al. (2011) found that a one degree Celsius temperature increase in Tanzania alone leads to an increase in cholera risk alone by 15 to 29%, and this does take into account potential effects from a simultaneous increase in rainfall variability. Moreover, there is currently a lack of understanding how the combination of multiple weather events and the interaction effects between rainfall and temperature affect waterborne disease incidence (Levy et al., 2018). Moore et al. (2017), for instance, found that El Niño events, which lead to a simultaneous increase in rainfall and warm weather across East Africa, leads to a threefold increase in Cholera outbreaks.

G.2 Climate change scenarios

Climate change impact analyses are typically based on so called climate change scenarios, which take into account projected future CO₂ concentrations and socioeconomic development, among other factors. In the IPCC Sixth Assessment Report by the UN Intergovernmental Panel on Climate Change (IPCC), climate change scenarios are represented by Shared Socio-economic Pathways (SSPs) (International Panel of Climate Change, 2022). The two scenarios most commonly used in impact analyses are the SSP2-4.5: “Middle of the Road” and SSP5-8.5: “Fossil-fueled Development”.

Each scenario assumes a trajectory taken in terms of sustainability measures. SSP2-4.5 assumes that greenhouse gas emissions remain at current levels as today until 2050, and then gradually fall towards 2100. This can be considered a relatively optimistic future scenario, though it may still lead to warming of up to about 2-3 degrees C relative to pre-industrial levels. The pessimistic scenario, which is commonly considered as an upper bound, is SSP5-8.5, which assumes very high greenhouse gas emissions, that triple by year 2075. This scenario may lead to rampant warming of up to almost 6 degrees C.

While our analysis is based on a continuous range of temperature increases instead of a fixed

interval, we highlight in our results in [Figure V](#) which temperature increase interval is associated with each scenario.

[[Table G.1](#) about here.]

G.3 The direct effects from temperature increases

Temperature is an important independent risk factor of waterborne diseases ([Levy et al., 2016](#)), and an essential condition for pathogen growth potential. To estimate the impact of a temperature increase on the effects of stagnant water, we use the results from a heterogeneity analysis where we split the data over +/-1 degree C temperature windows. This captures any non-linearity in the effect of temperature. We run this for the outcomes of diarrhea, test scores and school absence. The results of this analysis are reported in [Figure G.1](#).

While this analysis loses some power, we find suggestive evidence that effects increase with temperature consistent with the public health literature, and this trend is consistent across our three outcomes. Since diarrhea is the most salient outcome of the effects of waterborne disease, we use the heterogeneity results from these regressions as input to our impact analysis. Assuming a linear relationship, a regression run with the point estimates of the WBD coefficients for diarrhea gives a temperature slope coefficient of 0.18 with a p-value of 0.026. However, this does not take into account the uncertainty of each point estimate and may overstate the significance of this trend. To account for this, we instead run the trend regression weighted by the variance of each point estimate that takes into account the uncertainty of our temperature coefficients. This reduces the temperature slope coefficient to 0.11, with a p-value of 0.087. We consider this a more reliable and arguably more conservative estimate of the effects from temperature. In other words, this implies that a one degree C increase in temperature increases the WBD coefficient in the diarrhea regression by 0.11 units, which, relative to our baseline coefficient of 0.3, indicates a about a 30% increase in the effect on diarrhea per degree of warming, depending on assumed total warming.⁴⁶ This compares favorably to [Traerup et al. \(2011\)](#) who found an increase in Cholera incidence of up to 29% per each degree C of temperature increase.

[[Figure G.1](#) about here.]

⁴⁶To elaborate, one degree C of warming would increase the effect per degree C by 37%, while three degrees C of warming would increase the effect per degree C by 28%.

[Figure G.2 about here.]

G.4 The indirect effects from increases in rainfall variability

Climate change is expected to lead to a greater increase in rainfall variability, including an increase in intense rainfall events. This is because warm air enables more vapor and a greater evapotranspiration rates, which are the fundamental drivers of high-intensity convective rainfall (Pendergrass and Hartmann, 2014). This is the primary type of rainfall that drives our variation in stagnant water: when rainfall intensity exceeds soil infiltration capacity, this leads to surface runoff, which accumulates as stagnant water in low-lying areas. Hence, one way to estimate the impact of an increase in rainfall on stagnant water is to analyze how the frequency of intense rainfall events are expected to change with a warmer climate.

We use the latest projections from Li et al. (2021) from the latest CMIP6 Models⁴⁷ on how the frequency of rainfall events lasting 1 to 5 days with a current return period of 10 and 50 years⁴⁸ is expected to change with future temperature increases. Figure G.3 shows these relationships for East Africa specifically, where Tanzania is located.⁴⁹ They find that the frequency increases more or less exponentially with warming, and more so for the more intense 50-year rainfall events, for durations over both 1 and 5 days. Given the similarity in these increases for both return periods and durations, and given that these types of rainfall events are what are driving large increases in stagnant water, we use the estimated relationship in Figure G.3 to derive the effect of increases in rainfall variability on the baseline coefficient.

[Figure G.3 about here.]

G.5 Total effect

We then combine the direct effects from a temperature increase with the indirect effect temperature has on rainfall variability. The total burden of stagnant water can be considered as a function of the frequency with which exposure to stagnant water happens $F(T)$, which is a function of

⁴⁷This is an ensemble of simulations of future climate from the latest climate models that build on the SSP scenarios.

⁴⁸The return period is defined as the inverse of the annual probability of the event, i.e. a 10-year event has an annual probability of 10%.

⁴⁹There heterogeneity between regions is relatively small, and we would thus expect to find a similar increase in other locations subject to stagnant water and waterborne diseases.

temperature T , and the risk of an outbreak happening conditional on exposure to stagnant water, i.e. the magnitude of our baseline coefficient $M(T)$, also a function of T . Hence, a back-of-the-envelope estimation of the combination of these effects is simply: $F(T) \times M(T)$, which captures the multiplicative effects of a simultaneous increase in frequency and magnitude.

Figure G.4 shows the effect from each respective channel together with the total effect. The effect of temperature increases in a slightly more linear way than the effect of rainfall, but they are similar in magnitude over the temperature increases associated with the analyzed SSPs.

[Figure G.4 about here.]

We find that two degrees of warming leads to a threefold increase in the effect of stagnant water on waterborne disease outbreaks. This is identical in magnitude to the effect of El Niño events on Cholera in Tanzania, which are associated with simultaneous increases in rainfall and warm weather across East Africa (Moore et al., 2017).

G.6 Uncertainty and crucial assumptions

A complicating factor of climate change impact analysis is that society is expected to change across many dimensions other than just climate. By the end of this century, it is likely that developing countries such as Tanzania have seen an increase in economic growth and investments in public health and sanitation, factors that are likely to make the population more resilient to waterborne diseases. However, these developments are fundamentally uncertain, and we have thus abstracted away from them in the above analysis to arrive at the specific effects from climate change alone.

To get a better sense of how expected future conditions and ongoing trends may shape the waterborne diseases burden in the future, we summarize some of the relevant conditions in Table H.2, categorized by whether uncertainty is low, medium or high and whether it has a decreasing or increasing effect on the multiplier size. For example, economic growth is likely associated with urbanisation and population growth, and while economic growth itself may make households more resilient to disease through e.g. improved nutrition, population density itself, whether this is due to a greater overall population, or urbanisation, will likely make it easier for waterborne diseases to spread. This is especially the case if urbanisation increasingly happens on marginal land that

is at a higher risk of flooding and less connected to sewer networks that enable modern sanitation facilities.

The level of uncertainty is itself also of interest. For example, both climate scenarios and climate models are notoriously uncertain, and estimates for future temperatures differ widely depending on assumptions of future greenhouse gas emissions, and how the climate responds to these emissions. High uncertainty allows for small or insignificant changes in future climate, but in terms of social welfare, it likely only exacerbates social welfare. This is because, as [Figure V](#) shows, the effects of stagnant water on diarrhea is a *convex* function of temperature increases. The wider the temperature interval is, the greater is the expected cost from a temperature increase in terms of waterborne disease burden, which follows directly from Jensen's inequality. For example, while a scenario in which we know for certain that a future climate will be 2 degrees C warmer is expected to see a threefold increase in burden, a scenario in which we are equally likely to see warming over the span of 1 to 3 degrees C, the expected increase in the burden is about 3.4, more than 10% higher, and this only increases with our level of uncertainty.

In addition to uncertainty in future conditions that will affect how stagnant water affects waterborne diseases, an implicit assumption of our analysis is that there is no behavioral or physiological adaptation to an increase in waterborne disease risk. We believe this is still reasonable for two reasons. First, households today seem unaware of the risks associated with stagnant water, as discussed in [Section VII.B.](#), and hence we do not expect an increase in future risks to affect future awareness, *ceteris paribus*. Secondly, while immunity is typically developed after having survived exposure to bacteria and viruses, waterborne pathogens also include protozoa and parasitic worms, which are harder for the immune system to resist. Moreover, immunity against common viruses that cause stomach illness only tends to last a few months, and the sheer number and variation of different waterborne diseases occurring means that it is unlikely that any single individual develops immunity to the majority of these diseases. Lastly, the results of our analysis are consistent with the effects seen from El Niño events which recur with regular intervals. If there were behavioral and immune responses to these events in areas prone to stagnant water, we would not be likely to see the large increase in waterborne diseases that these events typically lead to.

[[Table G.2](#) about here.]

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Tables

Table I
Summary statistics: DHS and Uwezo

	Mean	SD	Min	Max
<i>Demographic and Health surveys</i>				
<i>A. Household and child characteristics</i>				
Birth Order	3.75	2.544	1	17
Girl	0.50	0.500	0	1
Age	1.93	1.416	0	4
Mother's age	29.3	7.108	15	49
Mother total fertility	4.12	2.588	1	17
Household wealth index	2.88	1.382	1	5
Sanitation - none	0.23	0.420	0	1
Sanitation - unimproved	0.46	0.498	0	1
Sanitation - shared	0.056	0.230	0	1
Sanitation - improved	0.26	0.438	0	1
<i>B. Cluster/Grid Cell characteristics</i>				
WDP [0.1]	0.017	0.0522	0	0.512
Local precipitation (cm)	0.0081	0.0101	0	0.223
Urban share of clusters	0.22	0.414	0	1
Num. villages/wave	493.6	146.5	176	608
Num. households/wave	324.4	21.98	303	351
Grid cells: 253 Obs: 21,471				
<i>Uwezo surveys</i>				
<i>A. Household and child characteristics</i>				
Test score (std)	-0.020	0.997	-4.481	4.239
Woman	0.46	0.499	0	1
Age	11.1	2.792	6	16
Mother's Age	36.3	5.903	16	60
Mother's Edu $\geq$ Sec	0.24	0.408	0	1
Wealth (index)	-0.0064	1.584	-2.230	24.58
Children in household	3.32	1.873	1	20
<i>B. Ward characteristics</i>				
WDP [0.1]	0.013	0.0361	0	0.812
Local precipitation (cm)	4.48	6.437	0	226.9
Rural ward	0.84	0.370	0	1
Ward area (sq.km.)	255.9	599.3	0.111	11437.0
Wards: 3876 Schools: 15,428 Obs: 386,005				

Note: Summary statistics of mean, standard deviation, minimum and maximum of each variable. Panel A displays statistics across individuals, while Panel B displays statistics at the local area level.

Table II
The effect of WDP on child health

	(1)	(2)	(3)	(4)	(5)	(6)
<i>Panel A. Health and physical outcomes</i>						
	Waterborne		Other and long run			
<i>Dep:</i>	Diarrhea	Weight.H.	Fever	Fever w/o diarrhea	Hemoglobin	Height
WBP	0.29*** (0.08)	-2.85 (3.27)	0.25** (0.10)	0.10 (0.10)	-5.84 (3.65)	3.51 (14.13)
Mean DV	0.11	98.07	0.18	0.13	105.07	90.20
Obs.	29,620	18,164	29,695	29,570	18,312	23,642
Clusters	272	252	272	272	266	272
<i>Panel B. Health and water</i>						
	Diarrhea		Fever without diarrhea			
	Not Breastfed	Breastfed	Not water	Water	Not water	Water
WBP	0.47*** (0.10)	0.16 (0.10)	0.04 (0.13)	0.29*** (0.09)	0.06 (0.20)	0.11 (0.12)
Mean DV	0.09	0.13	0.07	0.14	0.08	0.14
Obs.	12,372	17,246	4,120	15,712	4,112	15,696
Clusters	271	271	233	270	233	270

Note: Standard errors in parentheses clustered on DHS cluster/village level. WBD Potential is the share of area covered in stagnant water in the two weeks preceding date of survey. In panel A: Columns 1-2 displays outcomes plausibly affected by waterborne disease. In column 1, the dependent variable = 1 if child has suffered from diarrhoea the past two weeks, and column 2 weight for height as percentile of wave median. Columns 3-5 are additional outcomes of health issues less associated with waterborne disease: Fever, fever without diarrhea (to analyse fever as a separate symptom), hemoglobin (g/dl, a continuous measure of anemia risk) and height (cm). Diarrhea or fever = 1 if child has had in past two weeks). Cell, Wave, Calendar month fixed effects and sum of past two weeks precipitation used in all estimations. Individual level controls include birth order, multiple birth, gender, age, mother's age, total fertility of mother, toilet type. * p<0.1 ** p<0.05 *** p<0.01.

Table III
Effect of WDP on child mortality, by sanitation quality

	(1)	(2)	(3)	(4)
	<i>Dependent: Child died during treatment period (8 weeks)</i>			
	All	All	Urban	Rural
WDP	0.009 (0.010)	0.012 (0.014)	0.035** (0.017)	-0.008 (0.019)
Improved sanitation		0.002 (0.001)	0.004** (0.002)	0.000 (0.001)
WDP*Improved sanitation		-0.028 (0.027)	-0.042 (0.044)	-0.020 (0.023)
Mean DV	0.004	0.004	0.003	0.004
Avg % effect	2.6	-8.0	-14.1	-5.3
Obs.	32,215	24,542	6,627	17,894
Clusters	272	269	139	261

Note: This table reports the effect of WDP on child mortality in the past 8 weeks, interacted with a binary indicator representing access to improved sanitation (level 4 on the sanitation ladder). Standard errors in parentheses are clustered at the DHS grid-cell level. * p<0.1 ** p<0.05 *** p<0.01. WDP is the average share of the area covered by stagnant water in the eight weeks preceding the date of the survey.

Table IV
Effect of WDP on school absence

	School Absence			
	(1)	(2)	(3)	(4)
WDP	0.547*** (0.145)	0.599* (0.315)	1.919** (0.871)	0.268 (0.369)
WDP*Village sanitation		-0.148 (0.358)	-1.022 (0.669)	0.139 (0.524)
Village sanitation		-0.020 (0.012)	-0.022 (0.026)	-0.023 (0.015)
Sample	Full	Full	Urban	Rural
Ward FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Observations	65,053	65,053	16,378	48,675
Clusters	3,017	3,017	472	2,545

Note: This table reports the effect of WDP on school absence. Each observation is a class from grade 1 to grade 7. Standard errors in parentheses are clustered at the ward level. * $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$. WDP is the average share of ward covered in stagnant water in the eight weeks preceding the date of the survey. Covariates include average household wealth in the village and its interaction with WDP, a wealth index, local past eight weeks of ward precipitation. Ward is the treatment unit of observation. Wave is the year of the survey and waves included are 2011, 2013, 2014, 2015 and 2017. Month is the calendar month of treatment.

Table V
Effect of WDP on test scores

	Mean Test Score			
	(1)	(2)	(3)	(4)
WDP	−0.656** (0.287)	−1.503* (0.853)	−1.878 (1.784)	−1.331 (0.869)
WDP*Village sanitation		0.187 (1.093)	2.340 (1.443)	−0.876 (1.228)
Village sanitation		0.160*** (0.036)	−0.017 (0.078)	0.187*** (0.040)
Sample	Full	Full	Urban	Rural
Ward FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
HH controls	Yes	Yes	Yes	Yes
Observations	362,371	258,937	48,770	210,167
Clusters	3,834	3,198	487	2,711

Note: This table reports the effect of WDP on test scores. Standard errors in parentheses are clustered at the ward level. * $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$. WDP is the average share of ward covered in stagnant water in the eight weeks preceding the date of the survey. Covariates include child's gender and age, and mother's age and secondary education, a wealth index, local past eight weeks of ward precipitation. Ward is the treatment unit of observation. Wave is the year of the survey and waves included are 2011, 2013, 2014, 2015 and 2017. Month is the calendar month of treatment.

Table VI
Effects of WDP on waterborne disease awareness and preparedness

	(1) Diarrhoea important	(2) Breastfeeds	(3) Gave plain water	(4) Heard of ORS	(5) Used ORS	(6) Sanitation ladder
Panel A: Without gridcell FE						
WDP	−0.155** (0.064)	−0.198** (0.077)	0.027 (0.142)	0.165*** (0.039)	0.199*** (0.062)	−0.163 (0.252)
Observations	15,644	32,215	20,715	18,221	18,221	24,545
Clusters	242	272	271	242	242	272
Panel B: With gridcell FE						
WDP	−0.046 (0.041)	0.022 (0.076)	0.296*** (0.093)	0.063 (0.067)	−0.003 (0.145)	−0.024 (0.237)
Observations	15,644	32,215	20,714	18,221	18,221	24,542
Clusters	242	272	270	242	242	269

Note: Standard errors in parentheses clustered on DHS gridcell level. * $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$. WDP (%) is the average percent share of area covered in stagnant water the eight weeks prior to date of survey. 'Diarrhea importance' is an indicator for whether diarrhea is stated as the most important health problem in the community. 'Breastfeeds' is an indicator for whether the mother is currently breastfeeding. 'Gave plain water' is an indicator for whether a child received plain (untreated) water in the last 24 hours. 'Heard of ORS' is an indicator for whether a household heard about ORS, but never used it. 'Used ORS' is an indicator for whether a household used ORS. 'Sanitation ladder' is a sanitation quality index from 0-5.

Table A.1
Data sources used in the WDP algorithm

Input category	Dataset	Resolution and accuracy of input data
<i>Topography</i>	Digital Elevation Model from the Shuttle Radar Topography Mission	30 m horizontal resolution, aggregated into 90 m. 6 m vertical accuracy.
<i>Rainfall</i>	ERA5-reanalysis data for Precipitation	27 km horizontal resolution. 1 hour time resolution.
<i>Evaporation</i>	ERA5-reanalysis data for Potential Evapotranspiration	27 km horizontal resolution. 1 hour time resolution.
<i>Soil data</i>	ISRIC 2.0 global soil database	1 km horizontal resolution.

Table A.2
Validating WDB potential with observed surface water from satellite data

	All wards (1)	Switchers (5%) (2)	Switchers (10%) (3)
Panel A: All surface water (0-100% occurrence)			
WDP	0.154*** (0.018)	0.629*** (0.097)	0.370*** (0.090)
R ²	0.02	0.23	0.20
Observations	2,960	142	71
Panel B: Intermittent surface water (0-50% occurrence)			
WDP	0.064*** (0.004)	0.256*** (0.032)	0.259*** (0.038)
R ²	0.07	0.31	0.40
Observations	2,960	142	71
Panel C: Infrequent surface water (0-25% occurrence)			
WDP	0.027*** (0.002)	0.148*** (0.013)	0.109*** (0.013)
R ²	0.06	0.48	0.50
Observations	2,960	142	71

Note: *** p<0.01 ** p<0.05 * p<0.1. WDP is the eight-week average share of area of ward covered in stagnant water, $\sim(0,1)$ and the outcome is the average share of surface water in each ward over the period 1984-2021. Switchers (5%) and (10%) are the wards that switch from going to below 5% stagnant water to above, and from below 10% stagnant water to above, respectively, in the Uwezo sample. Panel A uses all observed surface water, including permanent water (100% occurrence), in the outcome variable. Panel B limits the outcome variable to only intermittent surface water (0-50% occurrence), while Panel C limits it to only infrequent surface water (0-25% occurrence).

Table A.3
Time to water by water source

	(1)	(2)	(3)	(4)
	<i>Dependent: Time to water (minutes)</i>			
	All	Tap	Well	Nature
WBD Potential	-7.675 (16.61)	-23.74 (29.35)	19.08 (31.65)	-61.45*** (22.17)
Mean DV	38	31	38	47
Obs.	22,686	5,851	8,507	3,824
Clusters	268	199	240	208

Note: Standard errors parentheses clustered on DHS grid-cell level. All estimations use calendar month, grid cell and wave fixed effects. *** $p < 0.01$ ** $p < 0.05$ * $p < 0.1$. WBD Potential is the average per cent share of the area covered in stagnant water the eight weeks prior to the date of survey. Each column represents a DiD estimation with a different subsample. The first column includes the full sample. We then divide the sample by which source of water the household states: Piped or tap water (2), from any type of well (3), from nature i.e. a river, dam, lake, stream, canal, pond (4). Individual level controls include birth order, multiple birth, gender, age, mother's age, total fertility of mother, toilet type.' Sum of past eight weeks precipitation per village/cluster also included.

Table B.1
Common waterborne diseases and symptoms

Pathogen type	Example	Common symptoms
<i>Bacteria</i>	Cholera, Salmonella (Typhoid fever), Shigella (Dysentery), E.coli, Legionella	Diarrhoea, Fever, blood in stool
<i>Viruses</i>	Rotavirus, Adenovirus, Astrovirus, Hepatitis A and E	Diarrhoea, Gastroenteritis, Fever
<i>Protozoa parasites</i>	Cryptosporidia, E. histolytica	Diarrhoea, Gastrointestinal illness
<i>Parasitic worms (Helminths)</i>	Roundworms, Hookworms, Trematodes (flat worms), Schistosomiasis	Fever, Abdominal pain, Diarrhoea, Gastrointestinal illness, Malnutrition,

Note: Sources: [Magana-Arachchi and Wanigatunge \(2020\)](#), [Hedley and Wani \(2015\)](#), [WHO \(2019\)](#).

Table E.1
Correlation between local precipitation and WBD Potential in the DHS sample

	WDP	Diarrhea	Diarrhea	Diarrhea
	(1)	(2)	(3)	(4)
WDP		0.290*** (0.077)		0.290*** (0.077)
Local rainfall	0.056 (0.081)		0.004 (0.123)	-0.015 (0.119)
Observations	32,215	29,620	29,620	29,620
Clusters	272	272	272	272

Note: Standard errors parentheses clustered on gridcell. Wave, calendar month and gridcell fixed effects included in all estimations. WBD Potential is the eight-week average share of area within a 10 km radius of the village covered by stagnant water, $\sim(0,1)$. 'Local rainfall' is the sum of rainfall in m over the past eight weeks, by DHS village and wave. *** $p < 0.01$ ** $p < 0.05$ * $p < 0.1$.

Table E.2
The effect of WDP on child health: additional outcomes

	(1)	(2)	(3)	(4)	(5)
	Diarrhea	Diarrhea fever	w/o Cough	Weight (kg)	Weight.A
WBP	0.290*** (0.077)	0.149*** (0.057)	0.135 (0.086)	-0.196 (0.506)	-5.732 (5.661)
Mean DV	0.11	0.05	0.18	10.92	88.49
Obs.	29,620	29,735	29,691	23,466	18,152
Clusters	272	272	272	272	252

Note: Standard errors in parentheses clustered on DHS cluster/village level. WBD Potential is the share of area covered in stagnant water in the two weeks preceding date of survey. Diarrhea not fever is the increase in cases of diarrhea without concurrent fever symptoms. Weight.A denotes weight-for-age in per cent of the reference median value. Cell, Wave, Calendar month fixed effects and sum of past two weeks precipitation used in all estimations. Individual level controls include birth order, multiple birth, gender, age, mother's age, total fertility of mother, toilet type. *** $p < 0.01$ ** $p < 0.05$ * $p < 0.1$.

Table E.3
Effect of WBD potential on diarrhoea: Urban-rural heterogeneity

	Urban	Rural
	(1)	(2)
WDP	0.209* (0.120)	0.266*** (0.101)
Observations	6,976	22,624
Clusters	140	264

Note: Standard errors in parentheses clustered on DHS gridcell level. All estimations use calendar month, gridcell and wave fixed effects and include individual covariates.*** p<0.01 ** p<0.05 * p<0.1. WBD Potential (%) is the average percent share of area covered in stagnant water the eight weeks prior to date of survey.

Table E.4
Effect of WBD potential on diarrhoea: Heterogeneity by gender

	Girls	Boys
	(1)	(2)
WDP	0.403*** (0.115)	0.160* (0.090)
Observations	14,770	14,848
Clusters	271	271

Note: Standard errors in parentheses clustered on DHS gridcell level. All estimations use calendar month, gridcell and wave fixed effects and include individual covariates.*** p<0.01 ** p<0.05 * p<0.1. WBD Potential (%) is the average percent share of area covered in stagnant water the eight weeks prior to date of survey.

Table E.5
Heterogeneity by access to improved water and sanitation

	(1)	(2)	(3)
	All	Urban	Rural
WDP	0.455** (0.202)	1.113*** (0.144)	0.439** (0.203)
Improved water	-0.014** (0.007)	0.007 (0.020)	-0.019** (0.008)
Unimproved sanitation (2)	-0.005 (0.010)	-0.007 (0.030)	0.003 (0.013)
Shared sanitation (3)	0.012 (0.015)	-0.002 (0.031)	0.029 (0.026)
Improved sanitation (4)	-0.002 (0.010)	-0.002 (0.032)	0.008 (0.012)
WDP * Improved water	0.040 (0.127)	-0.606*** (0.172)	0.055 (0.203)
WDP * Unimproved sanitation (2)	-0.214 (0.236)	-0.335* (0.189)	-0.233 (0.281)
WDP * Shared sanitation (3)	-0.085 (0.273)	-0.088 (0.375)	-0.386 (0.260)
WDP * Improved sanitation (4)	-0.588** (0.281)	-1.006*** (0.231)	-0.216 (0.474)
Observations	21,503	5,822	15,666
Clusters	268	139	260

Note: Standard errors in parentheses clustered on DHS gridcell level. All estimations use calendar month, gridcell and wave fixed effects.*** p<0.01 ** p<0.05 * p<0.1. WBD Potential (%) is the average percent share of area covered in stagnant water the eight weeks prior to date of survey. 'Shared toilet'=1 for households who use shared toilet facilities.

Table E.6
Shared toilets and the effect of WDP on diarrhoea

	(1)	(2)	(3)
	<i>Dependent: Child had diarrhea</i>		
	All	Urban	Rural
	<i>Panel A: Indicator for high-quality sanitation</i>		
WDP (0,1)	0.35*** (0.10)	0.31** (0.15)	0.30*** (0.10)
Improved sanitation	0.00 (0.00)	0.00 (0.01)	0.00 (0.01)
WDP*Improv. san.	-0.42** (0.20)	-0.77*** (0.22)	-0.00 (0.37)
Mean DV	0.11	0.13	0.11
Obs.	22511.0	6034.0	16462.0
Clusters	268	139	260
	<i>Panel B: Indicator for shared toilet</i>		
WDP	0.21** (0.10)	-0.08 (0.12)	0.24 (0.16)
Shared toilet	0.02*** (0.01)	0.00 (0.01)	0.02*** (0.01)
WDP*Shared toilet	0.06 (0.17)	0.37** (0.15)	-0.16 (0.19)
Mean DV	0.11	0.11	0.11
Obs.	21772.0	5864.0	15893.0
Clusters	260	132	253

Note: Standard errors in parentheses clustered on DHS gridcell level. All estimations use calendar month, gridcell and wave fixed effects.*** p<0.01 ** p<0.05 * p<0.1. WBD Potential (%) is the average percent share of area covered in stagnant water the eight weeks prior to date of survey. 'Shared toilet'=1 for households who use shared toilet facilities.

Table E.7
Shared toilets and the effect of WDP on child mortality

	(1)	(2)	(3)
	<i>Dependent: Child died in past 8 weeks</i>		
	All	Urban	Rural
	<i>Panel A: Indicator for high-quality sanitation</i>		
WDP (0,1)	0.012 (0.014)	0.035** (0.017)	-0.008 (0.019)
Improved sanitation	0.002 (0.001)	0.004** (0.002)	0.000 (0.001)
WDP*Improv. san.	-0.028 (0.027)	-0.042 (0.044)	-0.020 (0.023)
Mean DV	0.004	0.003	0.004
Obs.	24542.0	6627.0	17894.0
Clusters	269	139	261
	<i>Panel B: Indicator for shared toilet</i>		
WDP (0,1)	0.005 (0.014)	-0.024 (0.018)	0.009 (0.022)
Shared toilet	-0.002* (0.001)	-0.003* (0.001)	-0.000 (0.001)
WDP*Shared toilet	0.019 (0.024)	0.073*** (0.025)	-0.033 (0.024)
Mean DV	0.004	0.004	0.004
Obs.	23438.0	6381.0	17036.0
Clusters	260	132	253

Note: Standard errors in parentheses clustered on DHS gridcell level. All estimations use calendar month, gridcell and wave fixed effects.*** p<0.01 ** p<0.05 * p<0.1. WBD Potential (%) is the average percent share of area covered in stagnant water the eight weeks prior to date of survey. 'Shared toilet'=1 for households who use shared toilet facilities.

Table E.8
Effect of WBD Potential on diarrhoea: Sensitivity to control variables

	(1)	(2)	(3)	(4)	(5)	(6)
	<i>Dependent: Child has had diarrhoea</i>					
WBD Potential	0.293*** (0.0753)	0.277*** (0.0856)	0.276*** (0.0849)	0.290*** (0.0767)	0.280*** (0.0870)	0.290*** (0.0766)
Mean DV	0.11	0.11	0.11	0.11	0.11	0.11
Obs.	29,620	26,961	26,961	29,620	29,620	29,620
Clusters	272	266	266	272	272	272
Full sample:	✓			✓	✓	✓
Cov sample:		✓	✓			
Covariates:		✓		✓	✓	✓
Local precip:					✓	✓
Sampling weights:					✓	

Note: Standard errors in parentheses clustered on DHS gridcell level. All estimations use calendar month, gridcell and wave fixed effects. *** p<0.01 ** p<0.05 * p<0.1. WBD Potential is the average percent share of area covered in stagnant water the eight weeks prior to date of survey. Individual covariates include: Birth order, twin child, child sex, child age, mother's age, mother's total fertility, household toilet type, wealth index. Full sample includes all children's where the outcome, diarrhoea, is available. Covariate sample is the sample with all covariates – only toilet type and wealth index are ever missing. Column 1 is the specification with the full set of fixed effects but no control variables. In column 2 all individual covariates are included without accounting for the share of missing, and in column 3 we still limit the sample to children without controls but excluding covariates. In column 4 we account for missing variables by imputing an arbitrary number (-999) for missing variables and including a corresponding dummy variable = 1 for when the variable takes that value. Finally, in column 5 we include local precipitation as a control. Column 5 applies sampling weights and column 6 is our preferred specification without weights.

Table E.9
Effects of WBD potential on health outcomes without gridcell FE

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Waterborne		Other and long run				
	Diarrhoea	W.Age	Fever	Fever w/o di- arrhea	Cough	Hemoglobin	Height
WDP	0.300*** (0.062)	1.136 (3.512)	0.375*** (0.087)	0.231*** (0.059)	0.209*** (0.075)	-3.404 (4.069)	-3.419 (13.414)
Observations	29,620	18,152	29,695	29,570	29,691	18,312	23,642
Clusters	272	252	272	272	272	266	272

Note: Standard errors in parentheses clustered on DHS gridcell level. All estimations use calendar month and wave fixed effects and include individual covariates.*** p<0.01 ** p<0.05 * p<0.1. WBD Potential (%) is the average percent share of area covered in stagnant water the eight weeks prior to date of survey.

Table E.10
Effects of WBD potential on malaria awareness and preparedness

	(1) Malara imp.	(2) Avoid malaria	(3) Can protect	(4) Use bednet
Panel A: Without gridcell FE				
WDP	1.138*** (0.340)	0.351*** (0.091)	0.063** (0.030)	0.672*** (0.148)
Observations	15,644	10,233	32,215	21,948
Clusters	242	202	272	266
Panel B: With gridcell FE				
WDP	0.045 (0.256)	0.020 (0.123)	0.039 (0.028)	0.216 (0.149)
Observations	15,644	10,233	32,215	21,948
Clusters	242	202	272	266

Note: Standard errors in parentheses clustered on DHS gridcell level.*** p<0.01 ** p<0.05 * p<0.1. WBD Potential (%) is the average percent share of area covered in stagnant water the eight weeks prior to date of survey. 'Malaria important' is an indicator for whether malaria is stated as the most important health problem in the community. 'Avoid malaria' and 'Can protect' are indicators for whether an individual knows how to avoid contracting malaria and can protect themselves against malaria, respectively. 'Use bednet' is an indicator for whether an individual is currently using bednets against malaria.

Table F.1
Effect of WBD Potential on test scores: Additional specifications

	Dependent: Test Score (std)					
	(1)	(2)	(3)	(4)	(5)	(6)
WDP	-0.592** (0.272)	-0.654** (0.279)	-0.656** (0.287)	-0.663** (0.284)	-0.570* (0.304)	-0.598 (0.432)
Ward FE	✓	✓	✓	✓	✓	✓
Wave FE	✓	✓	✓	✓	✓	✓
Month FE	✓	✓	✓	✓	✓	✓
HH controls		✓	✓	✓	✓	✓
Rainfall control			✓	✓	✓	✓
Wave × Month FE				✓		
High WDP-specific linear trends					✓	
Ward-specific linear trends						✓
Observations	362,371	362,371	362,371	362,371	362,371	362,371
Clusters	3,834	3,834	3,834	3,834	3,834	3,834

Note: Standard errors in parentheses clustered on ward. WBD Potential is the average share of ward covered in stagnant water in the eight weeks preceding the date of the survey. HH controls include child's gender and age, and mother's age and secondary education, a wealth index, local past eight weeks of ward precipitation. Rainfall control is rainfall over the past 8 weeks. High WDP-specific linear trends is the interaction between an indicator for whether the average amount of stagnant water is 5% or more, and the continuous wave variable. Ward-specific linear trends is the interaction between the ward identifier and the continuous wave variable. Ward is the treatment unit of observation. Wave is the year of the survey and waves included are 2011, 2013, 2014, 2015 and 2017. Month is the calendar month of treatment. *** p<0.01 ** p<0.05 * p<0.1.

Table F.2
Effect of WBD Potential on test scores: By gender

	Male	Female
	(1)	(2)
WDP	−0.593* (0.332)	−0.632** (0.270)
Ward FE	Yes	Yes
Year FE	Yes	Yes
HH controls	Yes	Yes
Observations	194,608	167,748
Clusters	3,830	3,815

Note: The outcome is test scores. Standard errors in parentheses clustered on ward. WBD Potential is the average share of ward covered in stagnant water in the eight weeks preceding the date of the survey. Covariates include child's gender and age, and mother's age and secondary education, a wealth index, local past eight weeks of ward precipitation. Ward is the treatment unit of observation. Wave is the year of the survey and waves included are 2011, 2013, 2014, 2015 and 2017. Month is the calendar month of treatment. *** $p < 0.01$ ** $p < 0.05$ * $p < 0.1$

Table F.3
Effect of WBD Potential on test scores: By age

	Age 6-10	Age 11-16
	(1)	(2)
WDP	-0.628* (0.350)	-0.669** (0.290)
Ward FE	Yes	Yes
Year FE	Yes	Yes
HH controls	Yes	Yes
Observations	166,244	196,115
Clusters	3,828	3,827

Note: The outcome is test scores. Standard errors in parentheses clustered on ward. WBD Potential is the average share of ward covered in stagnant water in the eight weeks preceding the date of the survey. Covariates include child's gender and age, and mother's age and secondary education, a wealth index, local past eight weeks of ward precipitation. Ward is the treatment unit of observation. Wave is the year of the survey and waves included are 2011, 2013, 2014, 2015 and 2017. Month is the calendar month of treatment. *** p<0.01 ** p<0.05 * p<0.1

Table F.4
Effect of WBD Potential on absence: By grade

	Grades 1-3	Grades 4-7
	(1)	(2)
WDP	0.360** (0.164)	0.774*** (0.184)
Ward FE	Yes	Yes
Year FE	Yes	Yes
Observations	33,872	31,082
Clusters	2,945	2,914

Note: The outcome is test scores. Standard errors in parentheses clustered on ward. WBD Potential is the average share of ward covered in stagnant water in the eight weeks preceding the date of the survey. Covariates include child's gender and age, and mother's age and secondary education, a wealth index, local past eight weeks of ward precipitation. Ward is the treatment unit of observation. Wave is the year of the survey and waves included are 2011, 2013, 2014, 2015 and 2017. Month is the calendar month of treatment. *** p<0.01 ** p<0.05 * p<0.1

Table F.5
Effect of WBD Potential on test scores: By malaria prevalence in children

	Low Malaria	High Malaria
	(1)	(2)
WDP	−1.458*** (0.472)	−0.223 (0.406)
Ward FE	Yes	Yes
Year FE	Yes	Yes
HH controls	Yes	Yes
Observations	159,692	162,358
Clusters	1,515	1,610

Note: Results are the effect of WBD Potential on test scores. Wards are split by median Malaria mean prevalence in children aged 2-10 years in 2010 (approx. 7%). Standard errors in parentheses clustered on ward. *** $p < 0.01$ ** $p < 0.05$ * $p < 0.1$. WBD Potential is the average share of ward covered in stagnant water in the eight weeks preceding the date of the survey. Ward is the treatment unit of observation. Wave is the year of the survey and waves included are 2011, 2013, 2014, 2015 and 2017. Month is the calendar month of treatment.

Table F.6
Effect of WBD Potential on test scores: Robustness to heterogeneous and dynamic treatment effects

	TWFE	DCDH	DCDH, Switched in	DCDH, Switched out
	(1)	(2)	(3)	(4)
WBD $\geq 5\%$	-0.085** (0.041)	-0.055 (0.084)	-0.245*** (0.060)	0.024 (0.115)
Ward FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Observations	362,371	97,679	91,305	6,374

Note: Results are the effect of WBD Potential $\geq 5\%$ on test scores. TWFE is the baseline OLS model. DCDH implements the estimator robust to heterogeneous and dynamic treatment effects from (de Chaisemartin and D'Haultfoeuille, 2022). “Switched in” denotes wards that switched treatment status from untreated to treated between two consecutive waves, whereas “Switched out” denotes wards that switched from treated to untreated. Standard errors in parentheses clustered on ward. *** $p < 0.01$ ** $p < 0.05$ * $p < 0.1$. WBD Potential is the average share of ward covered in stagnant water in the eight weeks preceding the date of the survey. Ward is the treatment unit of observation. Wave is the year of the survey and waves included are 2011, 2013, 2014, 2015 and 2017.

Table G.1
Climate change scenarios used as references for the impact analysis

Scenario	Description	Avg. Temp. (C) Δ Pre-industrial	Freq 50-yr rainfall events Δ Pre-industrial
SSP2-4.5	<i>Middle of the road</i>	1.8-3.3	1.5-2.5
SSP5-8.5	<i>Fossil-Fuelled Development</i>	3.3-5.8	2.5-5.3

Note: SSPs includes projections of population growth, economic output and income available for adaptation.

Table G.2
Factors affecting the future health burden of stagnant water, categorized by the direction of their impact and their estimated uncertainty.

		EXPECTED IMPACT ON MULTIPLIER SIZE		
		Decreasing	Neutral	Increasing
UNCERTAINTY	High	• Medical innovation	• Behavioral adaptation	• Temperature • Rainfall variability
	Medium	• Improved water and sanitation • Public health investments • Economic growth		
	Low		• Immunity	• Population growth • Urbanisation

Figures

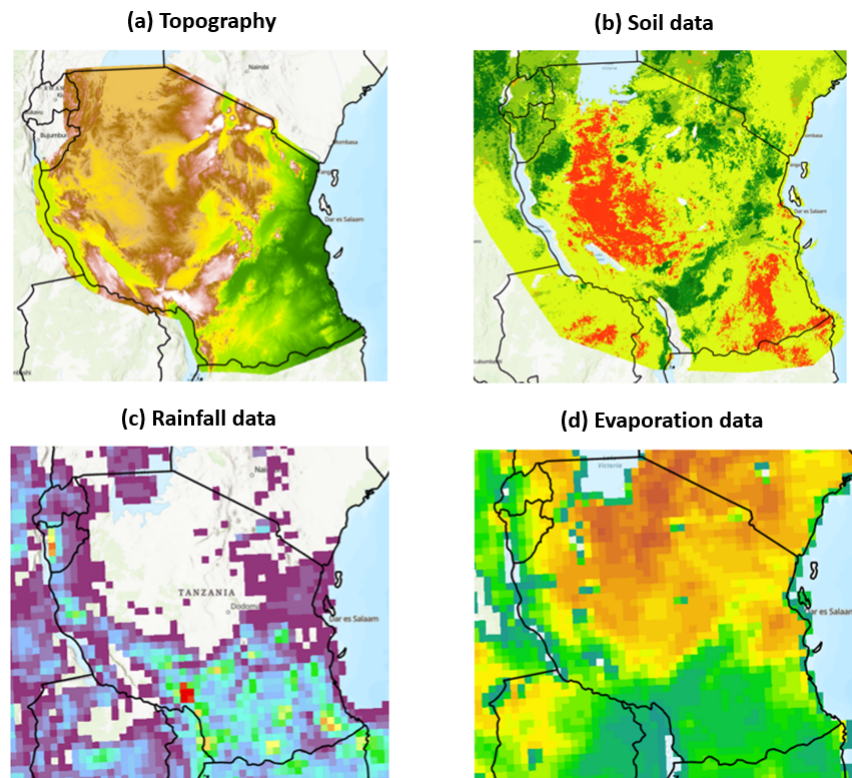


Figure I

Data sources used to construct the WDP measure

Note: The hydrological model uses as input four high-resolution gridded geographic datasets: (a) topographic data from SRTM, (b) soil infiltration data from the ISRIC 2.0 soil database, and (c) hourly rainfall and (d) potential evapotranspiration from ERA-5 reanalysis data.

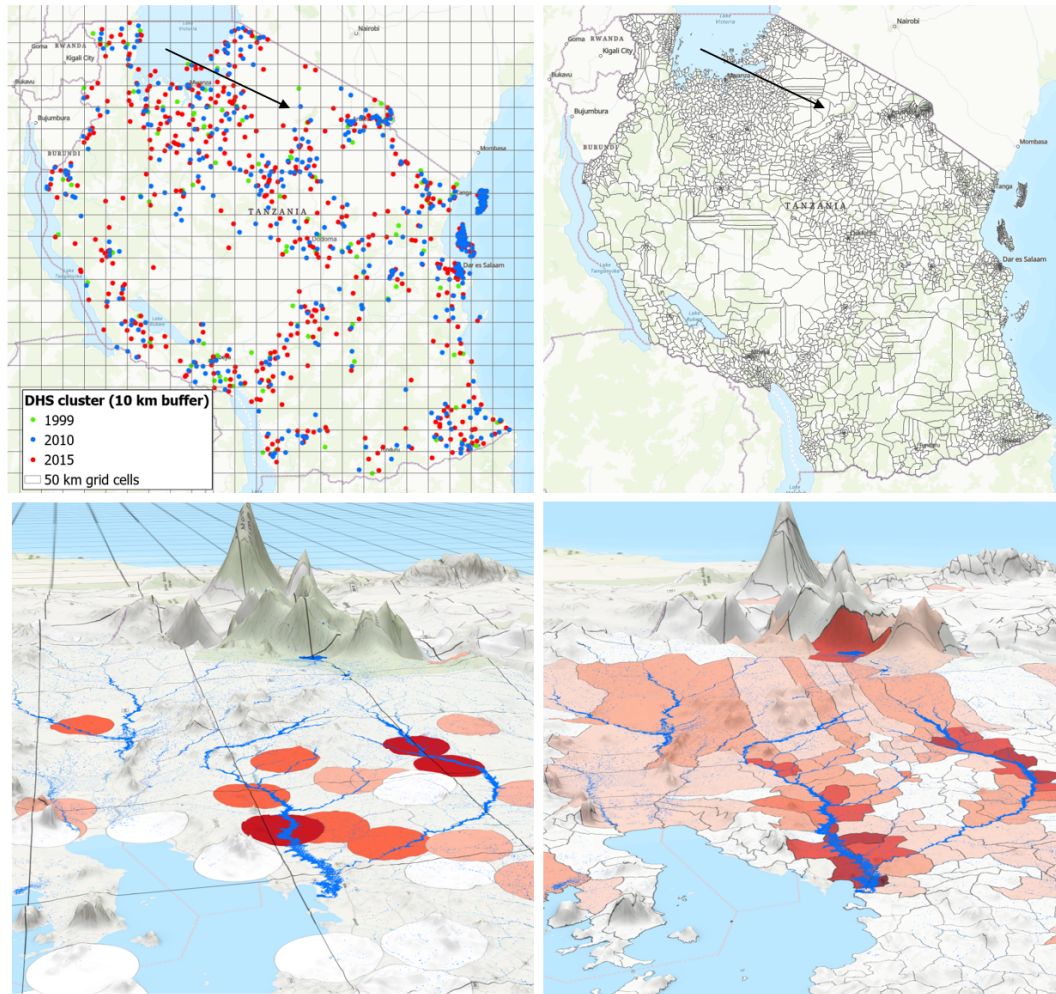


Figure II
WDP with DHS and Uwezo surveys

Note: This figure displays the output of our simulation which contains our main treatment variable, WDP, in one particular time step. Here we also highlight the difference in output between DHS (left) and Uwezo (right). In the top row, we display the available local area information we have. For DHS clusters we have different locations across waves thus necessitating the $50 \times 50 \text{ km}^2$ gridcells. For the Uwezo surveys, we have time-invariant wards. The bottom row shows the output for both types of spatial information for a subset of Tanzania (roughly indicated as the view along the black arrow with Lake Victoria towards the bottom of the figure). For DHS, treatment will be a weighted average of WDP intensity within the gridcell. For Uwezo, treatment is always given as a function of ward area.

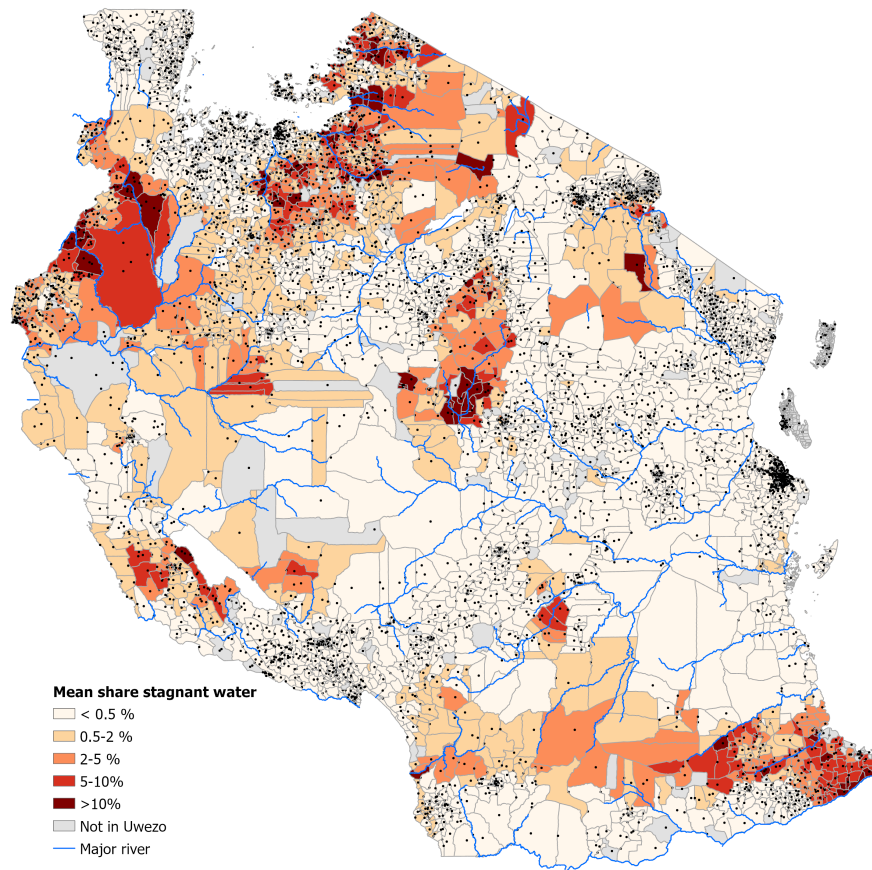


Figure III
Spatial distribution of WDP

Note: This map shows the geographic distribution of wards in Tanzania, together with population density (each dot represents 10,000 people), major rivers, and the mean share of stagnant water, our main treatment measure, in each ward in our sample period. Note that the mean share of stagnant water is only representative of the specific months and years represented in our Uwezo survey waves, which is the period for which we ran our hydrological model, and is not indicative of the long-term mean values for Tanzania.

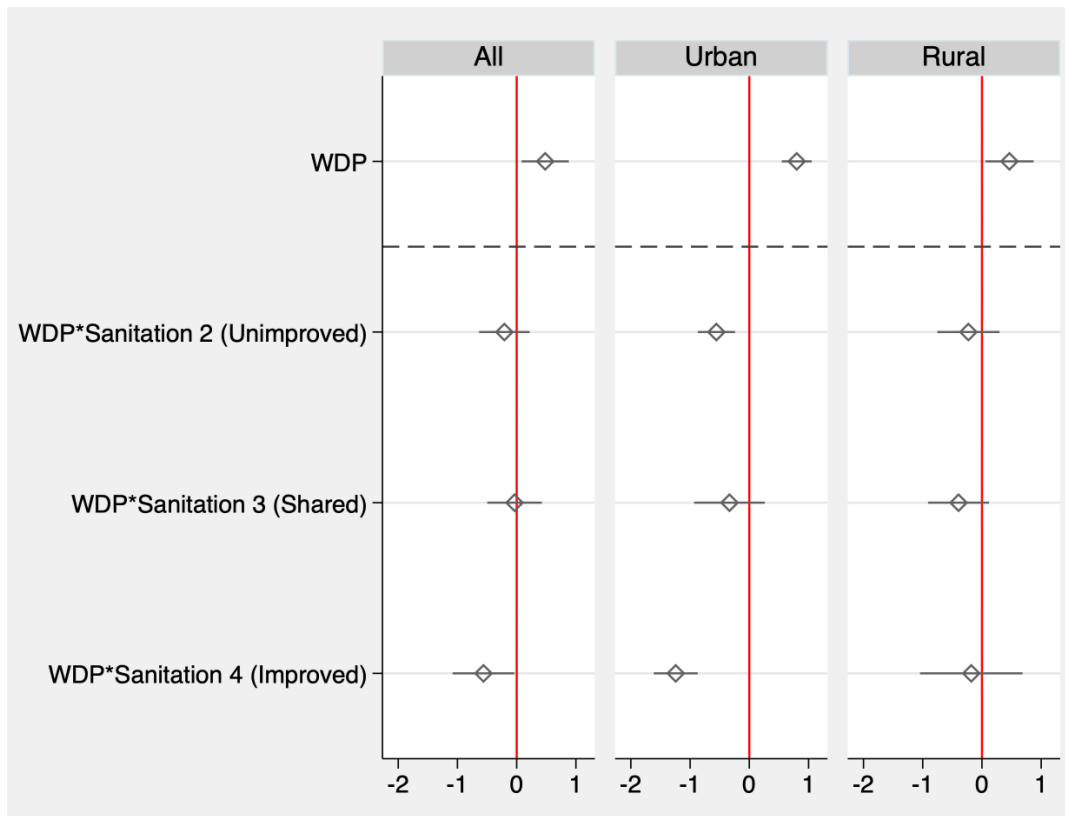


Figure IV
UN Sanitation ladder: Effect of WDP on diarrhea

Note: The three subgraphs represent separate triple-difference estimations of our main treatment, WDP, interacted with sanitation ladder and the effect on the probability the child has had diarrhea the past two weeks. The sanitation ladder is a categorical variable representing four types of sanitation facilities: No facilities, Unimproved sanitation, Shared facilities, Improved sanitation. Here, the baseline is households with the first sanitation ladder category: No facilities. The left subgraph presents the results using the whole sample of DHS children, while the remaining two divide the sample by whether the household lives in an urban or rural area (whether the DHS cluster is classified as urban or rural). See [Figure E.2](#) for more explanation of the sanitation categories.

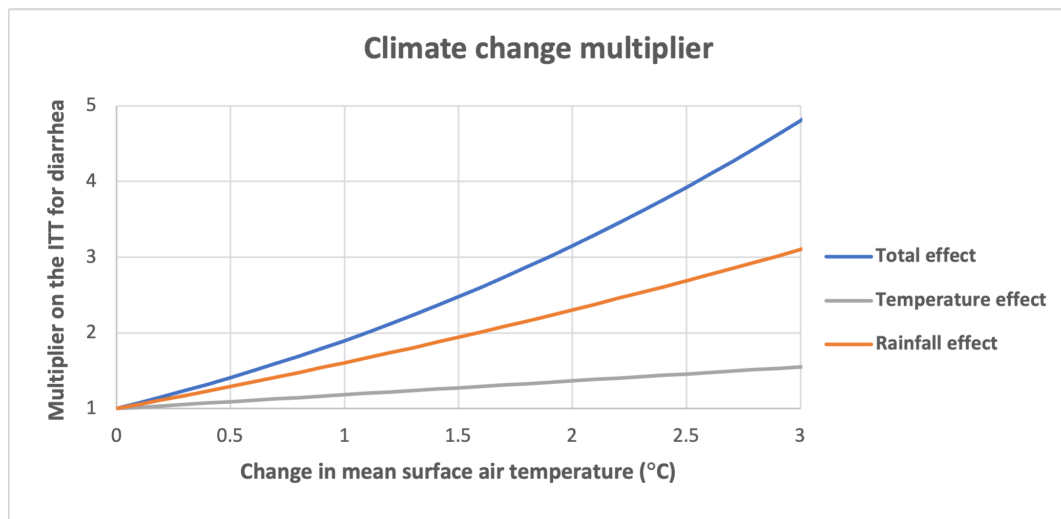


Figure V

Impact of climate change on the combined effect size and frequency of WDP on test scores

Note: This figure shows the combined increase in our baseline effect on test scores from the effect of an increase in temperature multiplied by the concomitant increase in frequency of intense rainfall events that drive stagnant water shocks. SSP 4.5 and SSP 8.5 constitute two Shared Socioeconomic Pathways (SSP) scenarios commonly used in climate change impact analysis.

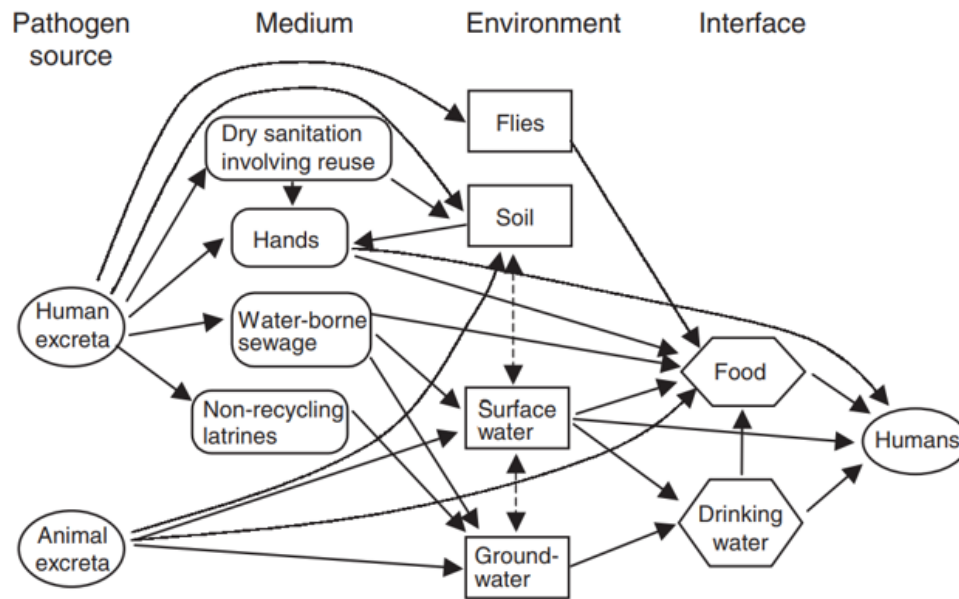


Figure B.1
Main transmission pathways of waterborne diseases

Note: This diagram shows the main transmission mechanisms for diarrhoeal diseases through the fecal-oral route, adapted from [Prüss-Üstün et al. \(2004\)](#).

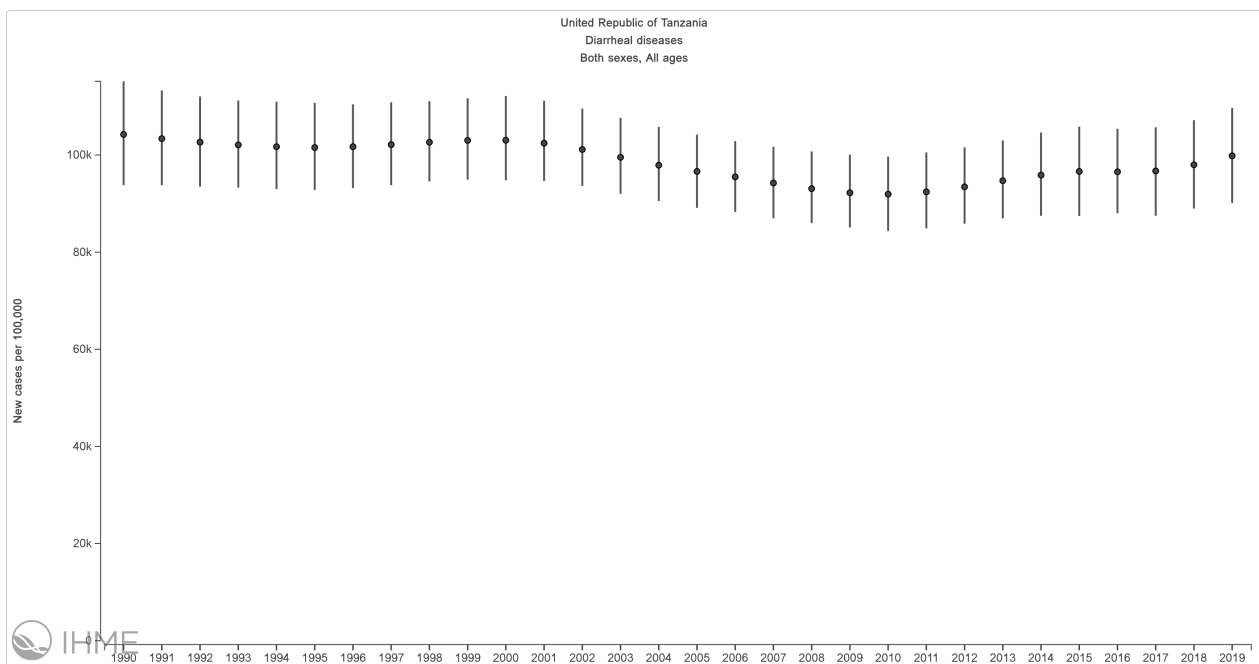


Figure B.2
Diarrhoea incidence by year in Tanzania

Note: This figure shows yearly diarrhoea incidence rates for Tanzania, based on data from the Global Health Data Exchange ([Global Burden of Disease Collaborative Network, 2021](#))

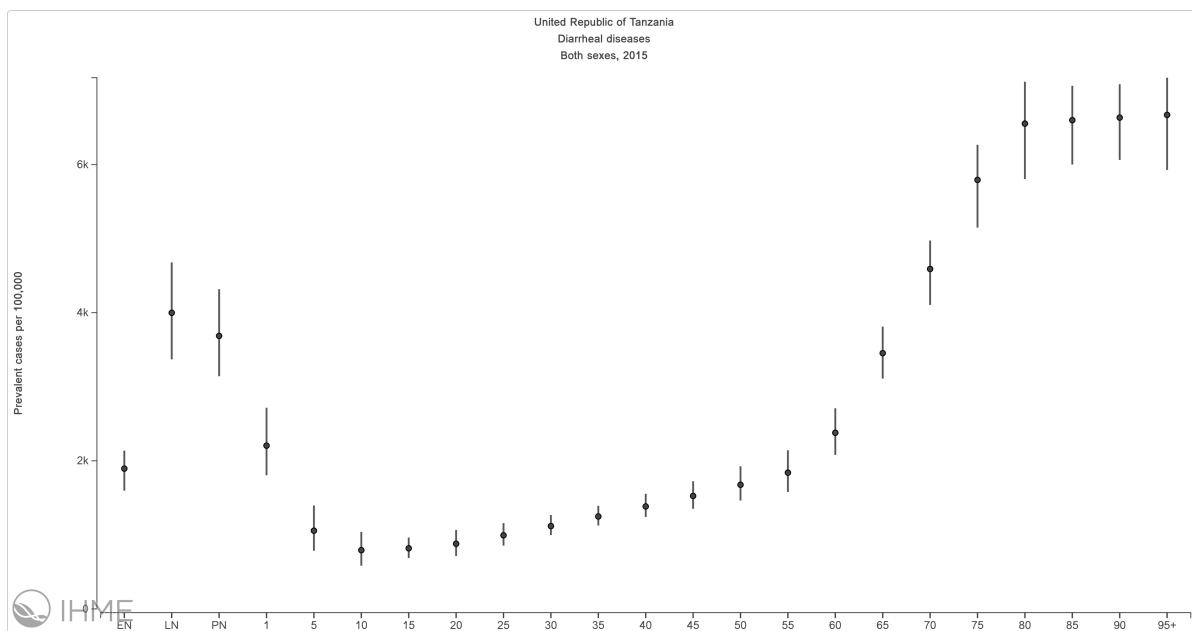


Figure B.3
Diarrhoea prevalence by age in Tanzania

Note: This figure shows diarrhoea prevalence rates for Tanzania, based on data from the Global Health Data Exchange ([Global Burden of Disease Collaborative Network, 2021](#))

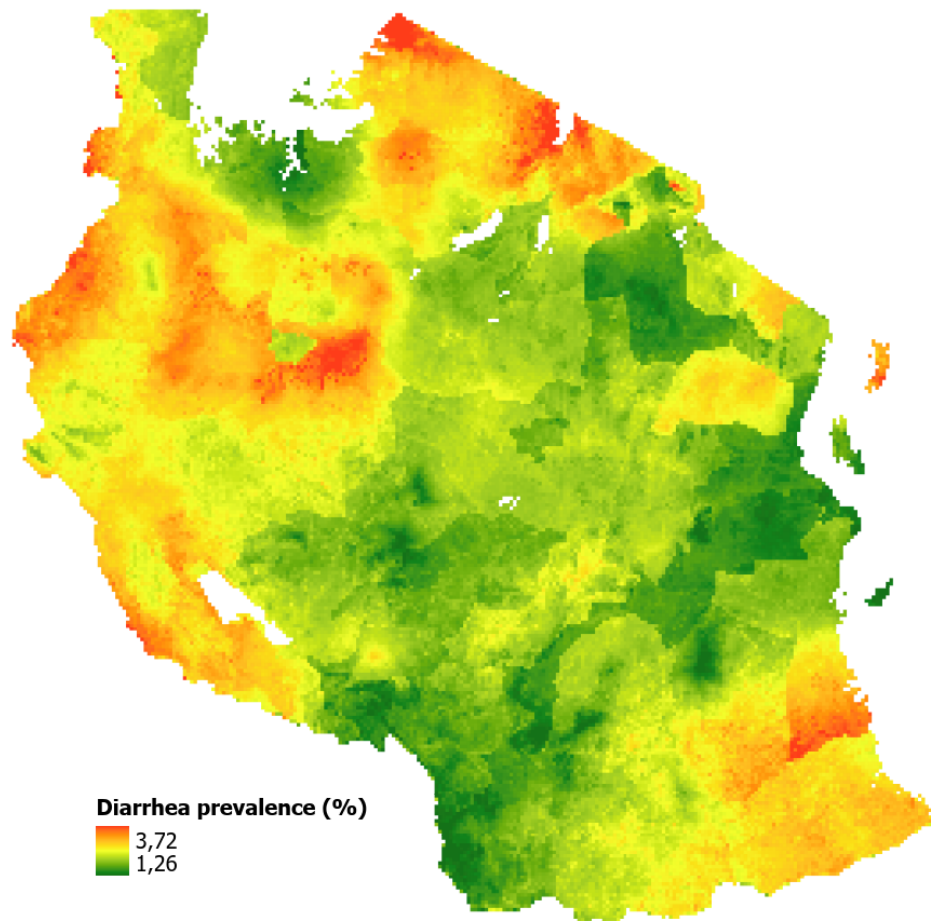


Figure B.4

Modelled geographic distribution of diarrhoeal disease in Tanzania

Note: This map shows the modelled geographic distribution of diarrhoeal prevalence for Tanzania, based on a Bayesian geostatistical model for diarrhea prevalence, from [Reiner et al. \(2020\)](#).

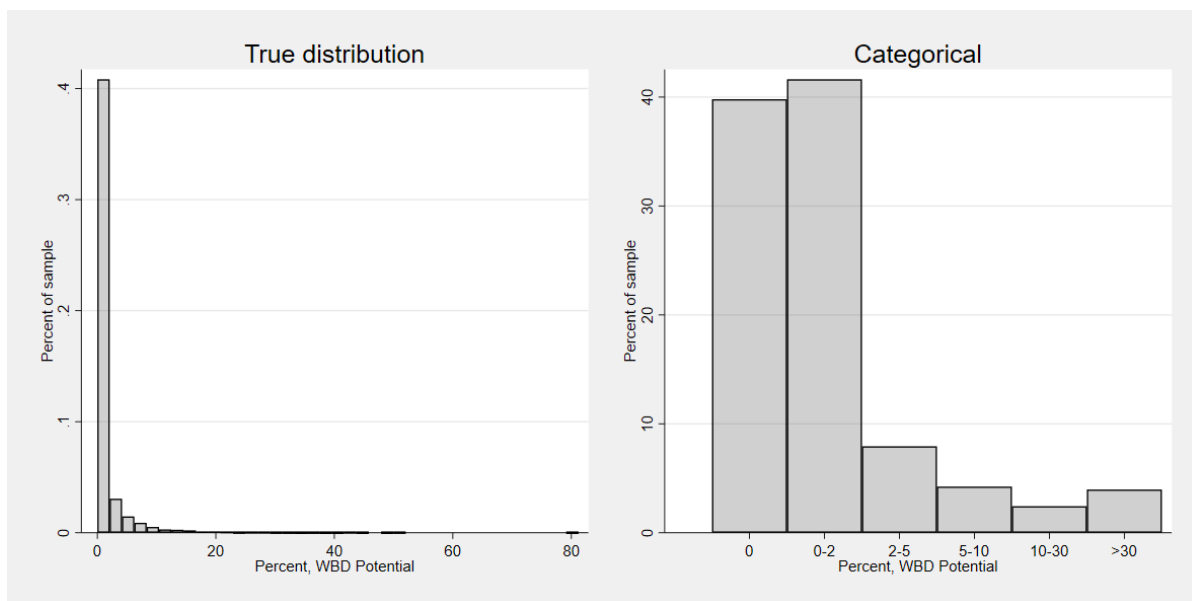


Figure D.1
Distribution of WBD Potential

Note: This figure depicts the distribution of waterborne disease potential (WBD Potential), where a ward w during wave and survey year y has one simulated value between 0 and 1, which is the share of the ward area covered by stagnant water. The left subplot depicts the distribution of WBD Potential, and to provide additional clarity the right subplot displays the distribution across five categories of shares of stagnant water.

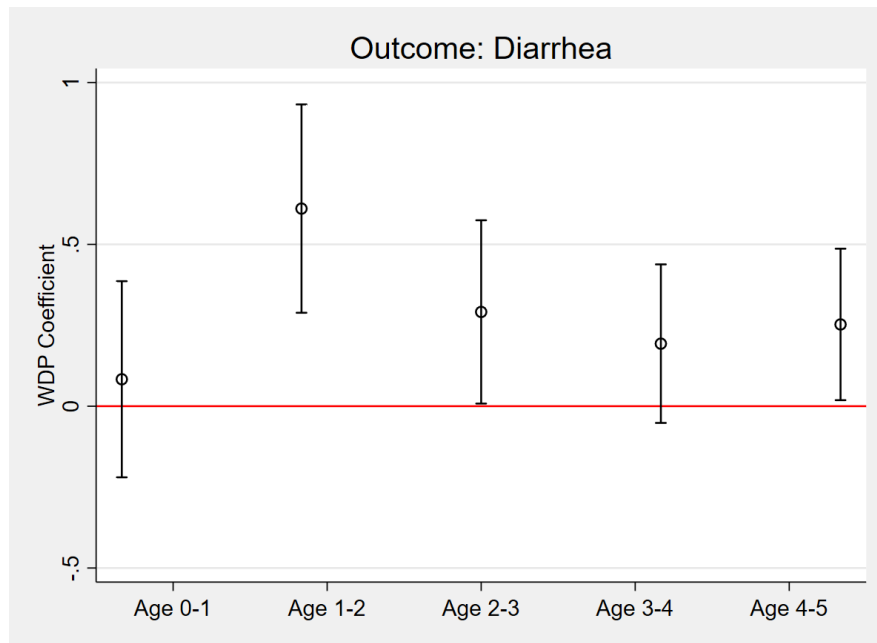


Figure E.1
WDP on diarrhea by age

Note: This figure summarises DiD estimate of WBD Potential on the share of children with recent diarrhoea. Coefficient and 95% confidence intervals are displayed. We run each regression separately for each age group. In all estimations we include calendar month, wave and ward fixed effects as well as individual and household covariates. Standard errors are clustered on ward-level.



Figure E.2
UN Sanitation ladder: Examples

Note: In this image we provide examples of how the different steps in the UN sanitation may look like. In reality, there are five steps of the ladder, the highest rung being modern sanitation. However, this category is not represented in our sample so for our case we consider a four-step ladder of sanitation. In 1, there are no facilities which includes open defecation. In 2, there are unimproved sanitation facilities that are not integrated to a well-functioning sanitation system. In 3, we include shared facilities. In 4, we have improved sanitation that are well-maintained and of higher standard and technology.

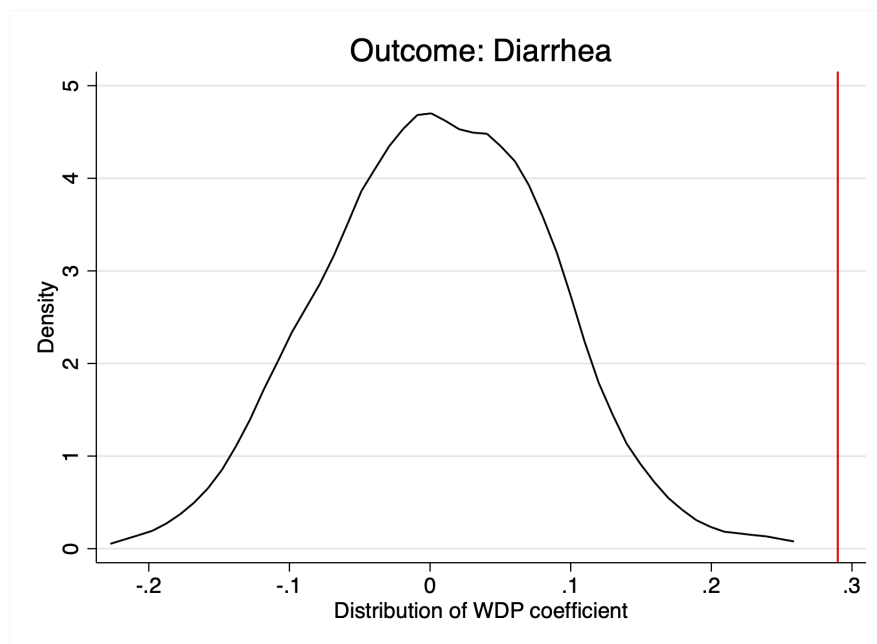


Figure E.3
Randomization Inference Test for WDP

Note: This figure shows the distribution of the WDP coefficient from 1000 regressions where in each regression WDP for a DHS village and wave is sampled with replacement from all DHS villages within the same gridcell. In all estimations we include calendar month, wave and ward fixed effects as well as individual and household covariates. Standard errors are clustered on ward-level.

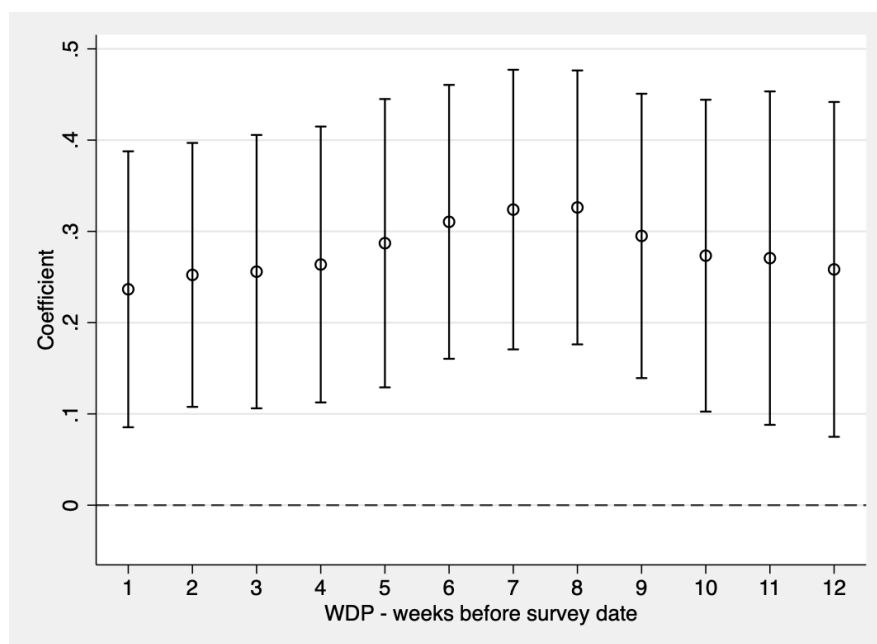


Figure E.4
WDP by week: Effect on diarrhoea

Note: This figure summarises DiD estimates of WBD Potential on diarrhea where each coefficient represents a regression with WDP defined as the share of stagnant water only within that particular week. Coefficients and 95% confidence intervals are displayed. In all estimations we include calendar month, wave and ward fixed effects as well as individual and household covariates. Standard errors are clustered on ward-level.

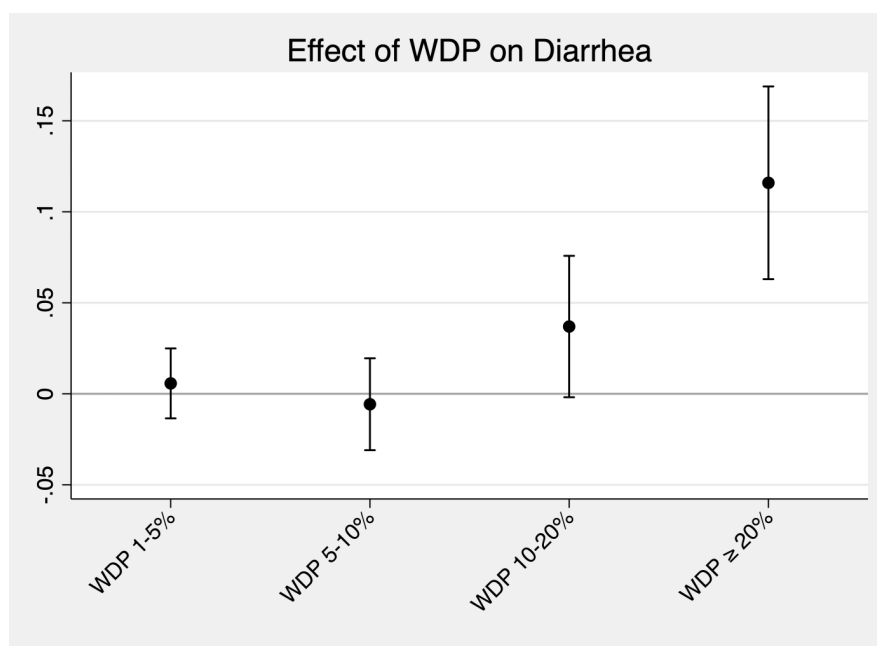


Figure E.5
WDP by binned magnitude: Effect on diarrhoea

Note: This figure summarises DiD estimate of WBD Potential on diarrhea. Coefficient and 95% confidence intervals are displayed. Here, we provide a series of dummies for different thresholds of WBD Potential. At baseline, WBD Potential is less than 1%. The first dummy variable thus captures the effect on diarrhea from WBD Potential between 1 and 5%. Calendar month, wave and gridcell fixed effects are included in the estimation as well as individual and household covariates. Standard errors are clustered on gridcell-level.

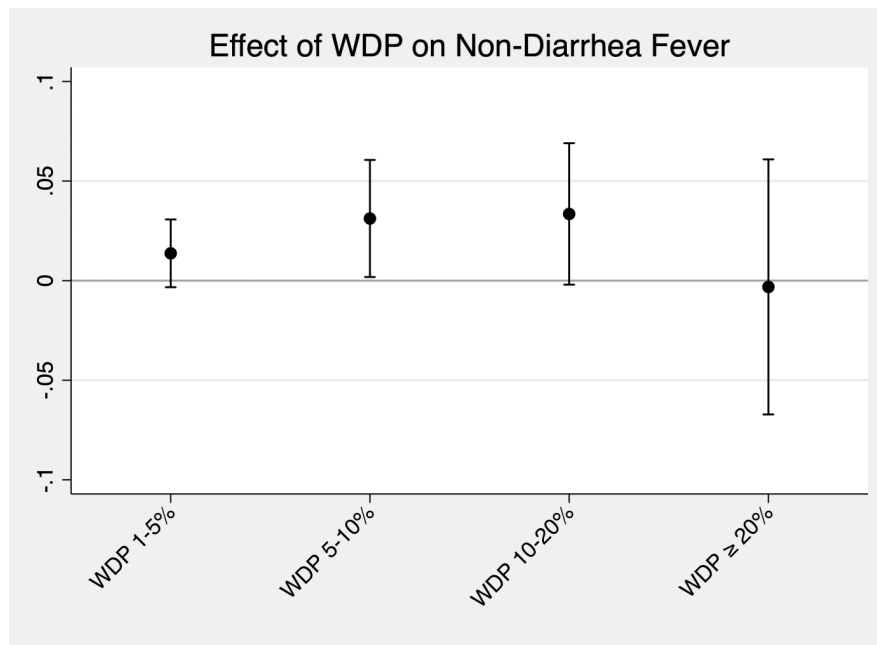


Figure E.6

WDP by binned magnitude: Effect on non-diarrhoea fever

Note: This figure summarises DiD estimate of WBD Potential on non-diarrhea fever. Coefficient and 95% confidence intervals are displayed. Here, we provide a series of dummies for different thresholds of WBD Potential. At baseline, WBD Potential is less than 1%. The first dummy variable thus captures the effect on non-diarrhea fever from WBD Potential between 1 and 5%. Calendar month, wave and gridcell fixed effects are included in the estimation as well as individual and household covariates. Standard errors are clustered on gridcell-level.

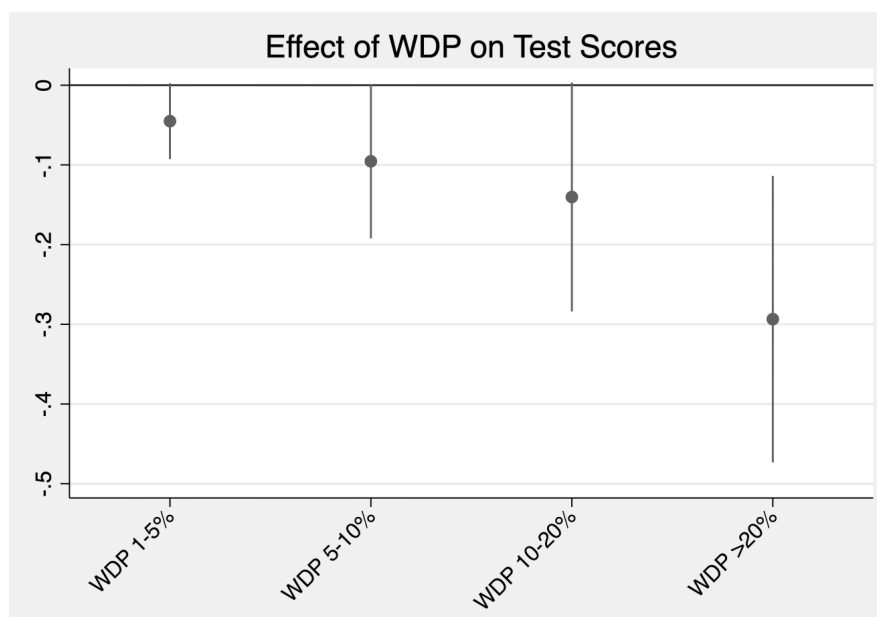


Figure F.1
WDP by binned magnitude: Effect on test scores

Note: This figure summarises DiD estimate of WBD Potential on test scores. Coefficient and 95% confidence intervals are displayed. Here, we provide a series of dummies for different thresholds of WBD Potential. At baseline, WBD Potential is less than 1%. The first dummy variable thus captures the effect on test scores from WBD Potential between 1 and 5%. Calendar month, wave and ward fixed effects are included in the estimation as well as individual and household covariates. Standard errors are clustered on ward-level.



Figure F.2

Randomisation test: Effect of WBD Potential on test scores

Note: This figure depicts the coefficient main DiD estimate capturing the effect of WBD Potential on test scores for 500 randomisation inference tests where WBD Potential has been permuted and the coefficients of WBD Potential on test scores summarised in a kernel density plot. Wave, Calendar month and Ward fixed effects and individual covariates are included in all estimations as well as individual covariates and standard errors are clustered by ward. 0.8% of the simulated coefficients are larger in magnitude in the negative direction than our estimate (indicated by the red line).

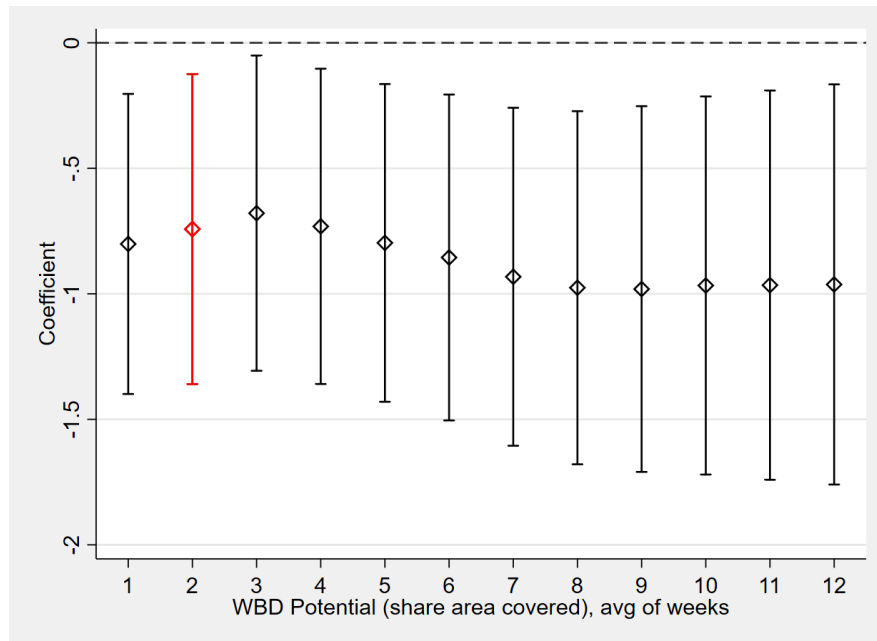


Figure F.3
Varying weeks in average measure of WDP

Note: This figure summarises DiD estimate of WBD Potential on test scores. Coefficient and 95% confidence intervals are displayed. Here, for each estimation we redefine the aggregation average of WBD Potential. In the first estimation, we define WBD Potential as the share of stagnant water in the one week preceding the date of survey when the child is tested. In the second estimate, we instead take the average share of stagnant water in the two weeks preceding the date of survey, and so on. The highlighted red estimate and confidence intervals reflect the definition we use in our main results, where we take the average of the eight weeks. In all estimations we include calendar month, wave and ward fixed effects as well as individual and household covariates. Standard errors are clustered on ward-level.

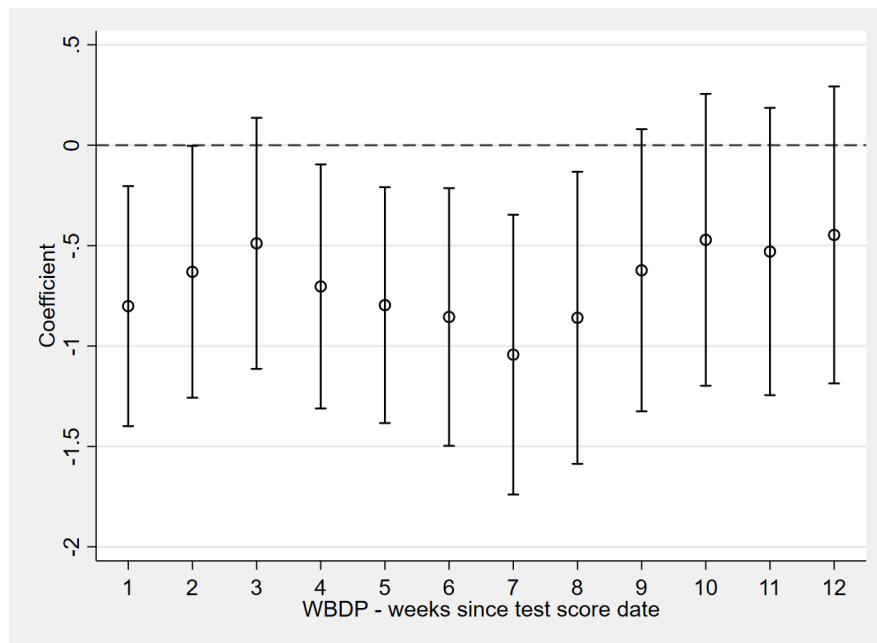


Figure F.4
WBD, by week discretely

Note: This figure summarises DiD estimate of WBD Potential on test scores. Coefficient and 95% confidence intervals are displayed. Here, for each estimation we redefine which week we include as the measure of WBD Potential. In the first estimation, we define WBD Potential as the share of stagnant water in the one week preceding the date of survey when the child is tested. In the second estimate, we instead take the share of stagnant water in the week which starts two weeks before the date of the survey and ends the week before the survey. Similarly, in the estimation labelled "3" we define WBD Potential as the share of stagnant water in (only) the week three weeks prior the date of survey. In all estimations we include calendar month, wave and ward fixed effects as well as individual and household covariates. Standard errors are clustered on ward-level.

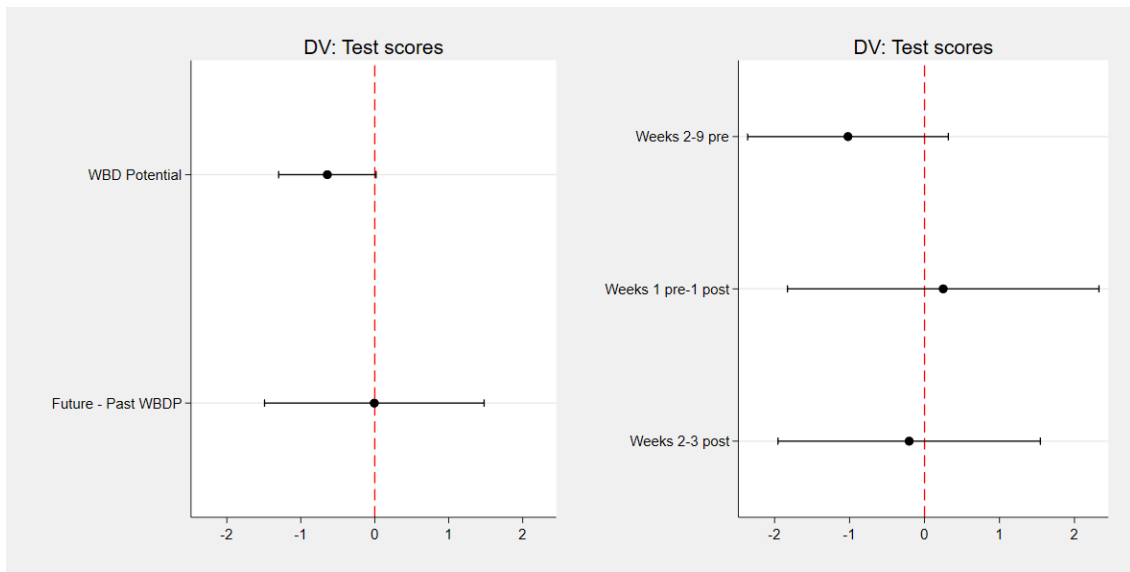


Figure F.5
Placebo: Effect of future WBD Potential

Note: This figure summarises DiD estimates of WBD Potential on test scores. Coefficient and 95% confidence intervals are displayed. Here, the two subplot represent one estimation each. On the left, we estimate with the main specification WBD Potential, and include the effect of the change in WBD Potential between the average stagnant water share *pre* survey, and the average stagnant water share *post* survey. On the right, we instead estimate longer-run means: Weeks 2-9 is the WBD Potential as the average 2-9 weeks prior to the survey. Weeks 1 pre- 1 post is the average WBD Potential the two weeks around the date of the survey, and the last estimate similarly is the average WBD Potential in the two weeks after the survey.

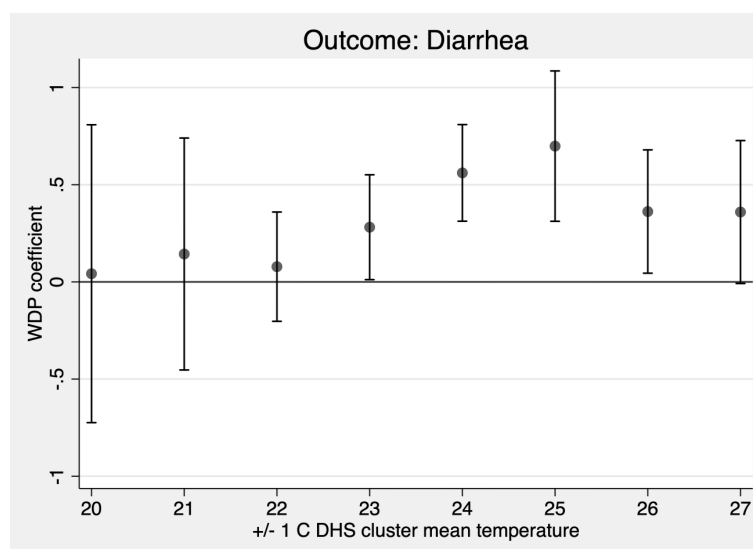


Figure G.1

Heterogeneity in the effect on diarrhea, using a 1 degree Celsius window

Note: This figure shows the effects of WDP on diarrhea, split by the average long-term temperature in the DHS cluster. Each point represents a regression run over a temperature window of +/- 1 degree Celsius. Coefficients and 95% confidence intervals are displayed.

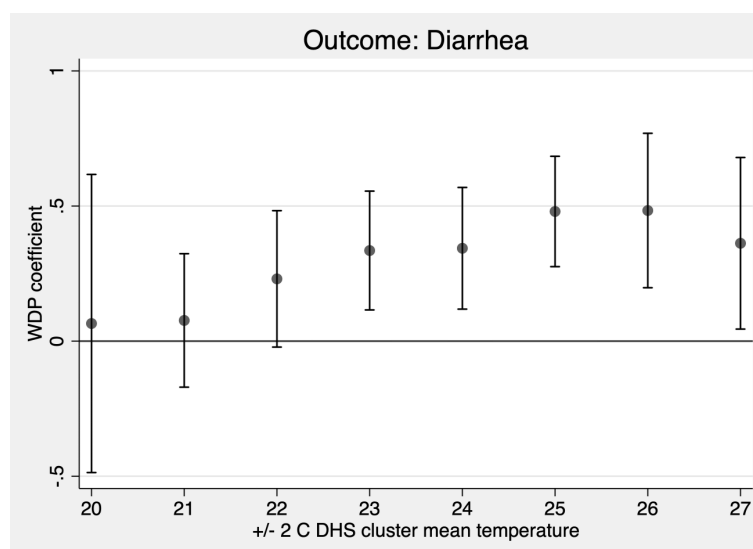


Figure G.2

Heterogeneity in the effect on diarrhea, using a 2 degree Celsius window

Note: This figure shows the effects of WDP on diarrhea, split by the average long-term temperature in the DHS cluster. Each point represents a regression run over a temperature window of +/- 2 degrees Celsius. Coefficients and 95% confidence intervals are displayed.

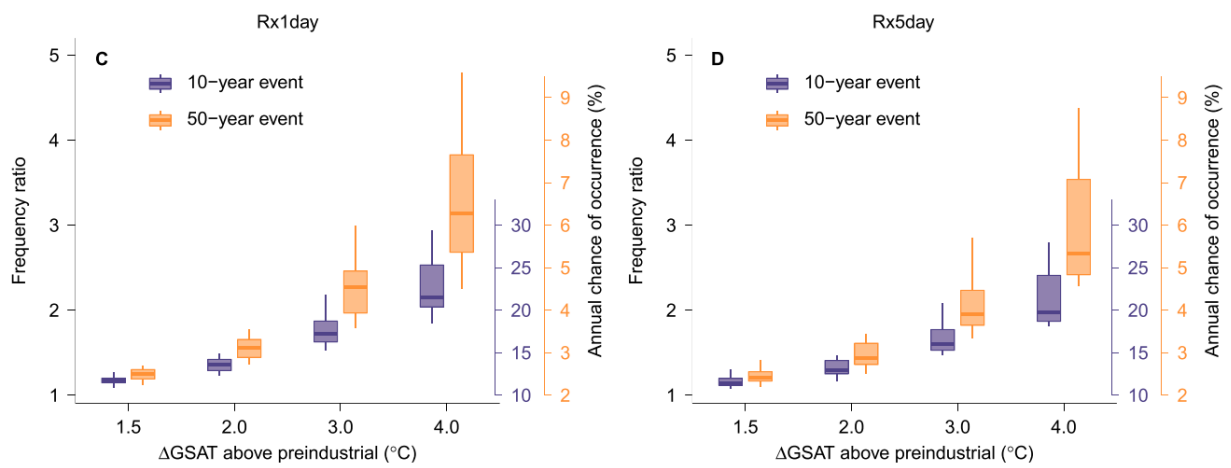


Figure G.3

Effects of global warming on the frequency of intense rainfall events

Note: From [Li et al. \(2021\)](#).

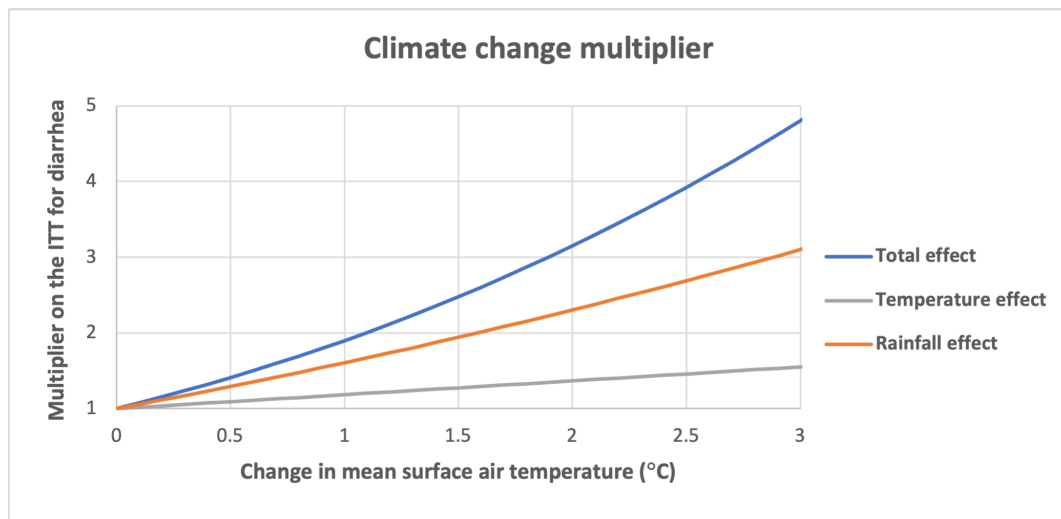


Figure G.4

Impact of climate change on the combined effect size and frequency of WDP on test scores

Note: This figure shows the combined increase in our baseline effect on test scores from the effect of an increase in temperature multiplied by the concomitant increase in frequency of intense rainfall events that drive stagnant water shocks. SSP 4.5 and SSP 8.5 constitute two Shared Socioeconomic Pathways (SSP) scenarios commonly used in climate change impact analysis.